# Medicaid Expansion and the Reallocation of Medical Residency Training*

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**Abstract**

We ask how the Affordable Care Act's Medicaid expansion changed where medical residents are trained. In an institution-level panel of the National Resident Matching Program linked to Medicare cost reports, expansion is followed by reallocation rather than contraction: intake at incumbent training institutions is roughly flat once panel entry and exit are coded correctly ($-1$ percent, not significant), while training capacity grows through newly entering institutions—many joining during the single-accreditation transition—so that state-level totals do not fall and, if anything, rise. This decomposition reconciles hospital-level and state-level evidence that previously pointed in opposite directions. An apparent role for state Medicaid GME financing formulas does not survive reclassification to the post-period formula rules and leave-one-state-out checks: the cross-formula contrast is a California phenomenon. We

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report inference candidly throughout; conservative standards distinguish no individual margin from zero.

**JEL**: I11, I13, I18, J44, H75

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# 1 Introduction

The geographic distribution of physicians shapes the population's health in measurable ways. A substantial body of evidence links primary care supply to improvements in life expectancy, disease-specific mortality, and preventable hospitalizations. Basu et al. (2019) estimate that each ten-physician increase in primary care supply per 100,000 population is associated with a 51.5-day gain in life expectancy and meaningful reductions in cardiovascular and cancer mortality. Macinko, Starfield, and Shi (2007) reach similar conclusions using state-level variation in primary care density, finding lower mortality rates for heart disease, cancer, and stroke in states with more primary care physicians. The relationship is not confined to the United States: evidence from Germany (Sundmacher and Busse 2011) and Canada (Piérard 2014) likewise documents health gains from improved access to general practitioners, suggesting the pattern reflects a general feature of primary care rather than an artifact of the US institutional context. Because where physicians complete residency training strongly predicts where they ultimately practice (LeFevre and Colwill 1983; Murphy 2020; Orr 2023), constraints on residency capacity translate into downstream constraints on the geographic distribution of physician supply. In this paper, we ask whether state-level Medicaid expansion changed the number of residency positions that hospitals offer and fill.

Whether policies designed to expand physician supply reliably produce these health benefits depends heavily on institutional context and program design. Evidence from earlier Medicaid expansions establishes that insurance coverage expansions can shift physician supply toward underserved areas through financial incentives. Qian (2024) show that 1980s expansions for pregnant women increased the number of OBGYNs in low-income counties, and Huh and Lin (2024) find that OBGYN counts per capita rose and remained persistently elevated following these expansions, with early-career physicians accounting for much of the growth. Parallel effects appear in dental care: Huh (2021) find that dental Medicaid expansions raised dentists per capita in poorer counties by 13 percent, and Buchmueller, Miller, and Vujicic (2016) document corresponding increases in dentist participation and patient access. Baker and Royalty (2000) further show that Medicaid-driven demand shocks were absorbed primarily by public hospitals and clinics rather than private practices, pointing to institutions as the key margin of adjustment. This last finding is particularly relevant to residency training, where hospitals both collect the expanded Medicaid revenue and decide how many training slots to fund.

International evidence introduces important qualifications. Carrillo and Feres (2019) evaluate a Brazilian initiative to increase primary care physician supply and find that while the program raised the volume of physician visits, it did not improve infant health outcomes, a pattern consistent with physicians substituting for nurses rather than generating new clinical value. A related evaluation of Brazil's More Doctors Program finds increases in healthcare utilization and reductions in preventable hospitalizations, though all-cause mortality remained unchanged (Mattos and Mazetto 2019). In Japan, Iizuka and Watanabe

(2016) show that a national residency reform intended to rationalize training pathways had the unintended consequence of drawing physicians away from rural areas, contributing to hospital closures as institutions faced higher wage pressures to attract and retain residents. Taken together, these experiences indicate that expanding physician training capacity is a necessary but not sufficient condition for improving health: the translation from training slots to health outcomes depends on where newly trained physicians choose to practice, the complementary inputs available in those settings, and the financial incentives governing provider behavior.

We answer this question using the residency programs that fill their positions through the national match. Each year, the National Resident Matching Program (NRMP) records the number of positions every program offers and the number it fills, which we link to the staggered timing of state Medicaid expansions under the Affordable Care Act. Because states expanded Medicaid in different years beginning in 2014, we estimate the effect with a staggered difference-in-differences event study, using the heterogeneity-robust imputation estimator of Borusyak, Jaravel, and Spiess (2024) and scaling matched positions by contemporary state population. Two features of the data construction turn out to be first-order, and we treat them as part of the paper's contribution. First, we build a crosswalk from NRMP sponsoring institutions to Medicare provider numbers, so that match intake and cost-report resident counts can be compared on the same institutions with the same weights. Second, we distinguish institution-years in which a sponsor was absent from the NRMP report from institution-years with genuinely zero positions—a distinction a balanced panel silently erases, and one that changes the answer.

Our central finding is reallocation, not contraction. On the balanced panel used in earlier drafts, expansion appears to lower matched positions by roughly 0.016 per 100,000 population, a 7.3 percent decline. But about a fifth of that panel's institution-years fall outside the window in which the institution appears in the NRMP report at all, and once those years are coded as missing rather than zero, the incumbent-institution effect shrinks to roughly $-1$ percent and is statistically indistinguishable from zero; restricting to institutions active in all ten years gives the same answer. Meanwhile, 223 of the 859 sponsoring institutions enter the NRMP during our window—many during the osteopathic single-accreditation transition—and they account for over 2,000 matched positions in expansion states by 2019. State-level training totals, which correctly include this entry margin, do not fall after expansion and, if anything, rise. The offered-position margin behaves like the matched margin throughout, so none of this reflects unfilled slots. Because treatment is assigned at the state level, few-cluster randomization inference, wild cluster bootstrap, and the Rambachan and Roth (2023) sensitivity analysis leave every individual estimate imprecise, and we report those standards uniformly; the decomposition itself—flat incumbents, growing entrants, non-declining state totals—is the robust object.

We examined—and ultimately reject as fragile—a financing mechanism that earlier drafts advanced. States whose Medicaid graduate medical education (GME) payments scale with patient volume had a direct financial incentive to expand slots when expansion raised Medicaid volume; states with fixed or no volume-based payments did not. Under the 2012 formula classification, the data appear to support this channel: positions fall by 11.7 percent in non-responsive states and not at all in volume-responsive ones (cross-group difference 0.035 per 100,000, $p < 0.001$). But our own classification file records that by 2015 New York, Illinois, and three smaller states had adopted volume-scaled formulas, so the 2012 vintage misclassifies the post period. Under the corrected classification the contrast shrinks by a third; dropping California—coded non-responsive because Medi-Cal pays no fee-for-service GME—flips its sign entirely, in the baseline design and in a design identified only from timing among adopters; and the payment-side differential in Medicare cost reports disappears ($p = 0.45$). We conclude that the cross-formula contrast is a California-versus-everyone comparison, and we report it as a candid negative result rather than a mechanism.

Our contribution is twofold. First, we provide an institution-level decomposition of how Medicaid expansion reshaped residency training—incumbent intake roughly flat, capacity growth through entry, state totals non-declining—which reconciles the hospital-level and state-level evidence that previously pointed in opposite directions, including the state-level aggregates in Ang and Beniwal (2026). Second, we build and release the data infrastructure that makes the decomposition possible: an NRMP-institution-to-Medicare-provider crosswalk, an entry-exit-corrected institution panel, and a two-vintage classification of state Medicaid GME formulas. Throughout, we bring modern few-cluster and pre-trend-sensitivity tools to bear and report what they do and do not support.

The remainder of the paper proceeds as follows. Section 2 describes the institutional setting; Section 3 develops the conceptual framework; Section 4 and Section 5 present the data and empirical strategy; Section 6 reports the main results and the financing mechanism; Section 7 presents robustness; and Section 9 concludes.

# 2 Background

Understanding how Medicaid expansion may have altered hospital training decisions requires engaging with three layers of institutional context. First, the expansion fundamentally changed the financial environment of teaching hospitals by converting uncompensated care into Medicaid revenue, but the magnitude and direction of that change varied across institutions depending on their pre-existing patient mix, their state's disproportionate share hospital (DSH) formula, and the design of state Medicaid graduate medical education (GME) funding. Second, residency training operates within a tightly regulated system of federal funding caps, accreditation requirements, and a centralized matching market, each of which shapes how hospitals translate financial incentives into slot decisions. Third, teaching hospitals are not a homogeneous group: they differ systematically in Medicaid exposure, ownership structure, training intensity, and geographic setting in ways that generate predictable heterogeneity in their responses to the same policy. This section develops each of these layers in turn.

## 2.1 The Affordable Care Act's Medicaid Expansion

Prior to the Affordable Care Act (ACA), approximately 35.6 million non-elderly adults in the United States lacked health insurance (Vistnes and Keenan 2019). Teaching and safety-net hospitals absorbed the resulting uncompensated care costs through commercial cross-subsidization and federal DSH payments, effectively operating as insurers of last resort at an estimated cost of roughly $800 per uninsured individual annually (Garthwaite, Gross, and Notowidigdo 2018; Lewin and Altman 2000).

The ACA extended Medicaid eligibility to nearly all adults with incomes at or below 138 percent of the federal poverty level. Following *NFIB v. Sebelius* (2012), adoption was voluntary, producing the staggered rollout that underlies the identification strategy in this paper. Twenty-five states and the District of Columbia expanded effective January 2014; additional states followed later in 2014 and in 2015, 2016, and 2019, while seventeen had not expanded as of 2019 (KFF 2026).[1] The federal government financed the newly eligible population at 100 percent through 2016, phasing down to a permanent 90 percent match by 2020, well above the standard federal medical assistance percentage (FMAP) of 57 to 63 percent. A large literature documents the resulting gains in coverage (Courtemanche et al. 2016; Glied and Jackson 2017; Kaestner et al. 2017), health outcomes (Courtemanche et al. 2018; Miller, Johnson, and Wherry 2021; Sommers et al. 2015; Sommers et al. 2017), and healthcare utilization (Courtemanche et al. 2017; Kobayashi et al. 2019; Nikpay et al. 2017). These studies focus on demand-side outcomes; the supply-side consequences for physician training capacity are the focus of this paper.

The hospital-level financial consequences of expansion are well documented. Dranove, Garthwaite, and Ody (2016) find that uncompensated care fell from 4.1 to 3.1 percent of operating expenses at hospitals in expansion states, with the largest reductions at safety-net institutions. Blavin (2016) finds corresponding improvements in operating margins. These revenue gains, however, exist in direct tension with the ACA's concurrent reductions to DSH allotments, which are calculated as a function of uncompensated care volume and decline automatically as the uninsured share falls. Critically, following *NFIB v. Sebelius*, hospitals in non-expansion states faced mandated DSH cuts while continuing to absorb high uncompensated care volumes, creating a pronounced fiscal asymmetry (Linehan 2013; Ossei-Owusu 2017). The net revenue effect of expansion at any given hospital therefore depends on its pre-expansion patient mix, its state's DSH formula, and the generosity of state Medicaid GME funding.

The ACA also incorporated targeted physician workforce provisions. Heisler (2013)

[1] Several states adopted expansion after 2019, but this paper focuses on the 2010 to 2019 period, which captures the first five post-expansion years for early adopters and provides a balanced pre- and post-expansion window for identification.

catalogs these, including Medicare payment adjustments, new GME funding streams, and the redistribution of unused Medicare residency slots under Section 5503 to hospitals in areas with documented physician shortages, with a statutory requirement that 75 percent of reallocated slots support primary care or general surgery training. McNamara and Hussain (2025) find that these cap changes increased program size, and McNamara and Pineda-Torres (2025) show that Section 5503 raised primary care physician supply by 4.1 percent in treated areas. These GME-targeted interventions confirm that the legislative environment surrounding Medicaid expansion was actively attempting to expand the training pipeline, making it essential to isolate any effect attributable to the Medicaid coverage channel specifically.

Evidence from earlier public insurance expansions establishes that coverage expansions can shift physician supply through institutional channels. Garthwaite (2012) shows that the CHIP expansion changed physician behavior on both the participation and hours margins—the closest antecedent to our question of how a coverage expansion feeds back into the supply side of medicine—and Barnes et al. (2023) document that public insurance expansions raise labor demand in physician practices. Qian (2024) shows that 1980s Medicaid expansions for pregnant women increased OBGYN counts in low-income counties, and Huh and Lin (2024) find these effects were persistent and driven by early-career physicians. Baker and Royalty (2000) shows that Medicaid demand shocks were absorbed primarily through public hospitals and clinics rather than private practices, a pattern directly relevant here given that hospitals control residency slot capacity. For the ACA expansion itself, Neprash et al. (2021) find increases in Medicaid participation and payer-mix shifts in primary care with little change in physician labor supply, and Candon et al. (2018) document that appointment availability for Medicaid patients tracks fee levels—both underscoring that the accommodation of new coverage runs through institutions rather than individual physicians. Because Medicaid pays physicians well below Medicare in most states (Zuckerman, Skopec, and Liang 2021), the revenue consequences of expansion for teaching hospitals depend on volume as much as price, which is why the financing formulas described below matter.

## 2.2 The US Physician Training System

### 2.2.1 Residency Training and the Match

Residency programs provide specialty-specific supervised clinical training and are a prerequisite for board certification and independent licensure in all specialties. Programs are accredited by the Accreditation Council for Graduate Medical Education (ACGME) and vary in length from three years in family medicine, internal medicine, and pediatrics to five years in general surgery. Any constraint on residency capacity translates directly into a constraint on future physician supply, and where physicians train strongly predicts where they ultimately practice (LeFevre and Colwill 1983; Murphy 2020; Orr 2023), making hospital-level training decisions a first-order determinant of long-run physician workforce distribution. Entry into residency is coordinated through the National Resident Matching Program (NRMP) Main Residency Match, a centralized clearinghouse in which medical school graduates and programs submit rank-ordered preference lists that are resolved by a deferred acceptance algorithm (Gale and Shapley 1962; Roth and Peranson 1999); Agarwal (2015) estimates an empirical model of this market and shows that program capacities, not applicant scarcity, bind for most programs. Approximately 40,000 positions are offered annually across more than 6,000 programs.

### 2.2.2 Creating and Expanding Residency Programs

Developing a new residency program requires substantial institutional commitment well before the first resident class. Barajaz and Turner (2016) describe four overlapping phases: infrastructure development, educational design, ACGME accreditation, and recruitment. Programs must demonstrate sufficient physical, human, and financial resources, and ACGME approval is not automatic (Accreditation Council for Graduate Medical Education 2022). These preparatory investments represent sunk costs committed years in advance, making new program creation unlikely in response to anything other than a durable change in the financial environment.

Expansion of existing programs is constrained by the Balanced Budget Act of 1997, which fixed each hospital's Medicare-funded resident count at its 1996 level. Training residents above this cap requires hospitals to finance the full marginal cost from operating budgets, covering salaries, benefits, and faculty supervision time. Medicaid GME add-on payments partially offset this cost in states with volume-responsive formulas, but above-cap expansion is generally a discretionary expenditure sensitive to operating margins. Program complements introduce a further constraint: each program's maximum resident count is set by the ACGME based on faculty availability and clinical caseload, and increasing a complement requires formal approval (Accreditation Council for Graduate Medical Education 2022). Financial capacity and faculty availability jointly determine whether a hospital can meet the conditions for growth even when the regulatory pathway exists.

### 2.2.3 GME Funding

GME in the United States is financed through a multi-payer system including Medicare, Medicaid, the Health Resources and Services Administration (HRSA), the Veterans Health Administration (VHA), and the Department of Defense (DoD) (He, Whang, and Kristo 2021). Medicare is the dominant source, providing Direct (DGME) and Indirect Medical Education (IME) payments totaling approximately $16.2 billion in fiscal year 2020 (Congressional Research Service 2022); whether these subsidies determine the level of training, or instead accrue to hospitals training residents they would fund anyway, is a long-standing question (Newhouse and Wilensky 2001; Nicholson 2002), and Medicare

Payment Advisory Commission (2023) reviews the current policy debate. Its payment *formula* is fixed in federal statute, so Medicaid expansion does not change Medicare GME rules directly; realized Medicare payments can nonetheless move with a hospital's resident counts and the patient-share terms of the formula, which is why we treat measured payments as an equilibrium outcome rather than a mechanical transfer when we use them as evidence on the financing channel (Section 6.3). Medicaid GME ranks second at $7.39 billion in 2022, up from $3.78 billion in 2009 (Henderson 2021, 2023), but unlike Medicare it is entirely state-determined. States vary substantially in whether they fund GME at all, in payment generosity, and in formula design (Eden, Berwick, and Wilensky 2014; Henderson 2013, 2016, 2021, 2023; National Conference of State Legislatures 2024). Some operate volume-responsive systems that scale payments with Medicaid patient volume; others use fixed per-resident amounts, managed care capitation rates, or direct grants. Federal matching amplifies all state payments, so state-level formula design choices have outsized consequences for teaching hospital resources. In states with volume-responsive formulas, expansion mechanically raises Medicaid patient volume and triggers GME revenue increases without additional legislation. In states with fixed formulas, the same volume gain generates no incremental GME revenue. This cross-state variation is a key source of identifying variation in the mechanism analysis. HRSA's Children's Hospital GME and Teaching Health Center GME programs provide additional targeted support for pediatric and primary care training, insulating those programs from the operating margin pressures that affect procedurally intensive specialties (He, Whang, and Kristo 2021).

## 2.3 Hospital Heterogeneity and Differential Medicaid Exposure

A companion paper by one of us, Ang and Beniwal (2026), studies the same policy at the state level and reaches what initially looks like the opposite conclusion: that expansion *increased* GME funding and expanded training capacity, particularly at urban hospitals. The two papers differ in unit of analysis (state aggregates versus NRMP sponsoring institutions), outcome construction, and estimator, and earlier drafts of this paper papered over the tension with the word "complementing." We now believe the two findings are the two halves of one decomposition: state aggregates include the entry margin, where capacity grew, while a balanced institution panel isolates incumbents, whose intake was roughly flat. Section 6.2 makes this reconciliation explicit, and the concluding section states which estimate a policymaker should use for which question.[2] Whether aggregate effects reflect concentrated responses at specific institutions or diffuse adjustments across the population is exactly the question the decomposition answers. This section characterizes

[2]The companion paper's outcome definition, data source, sample and years, estimator, and point estimate with its standard error are summarized alongside ours in Section 6.2. [The companion draft's headline estimate and standard error are to be inserted from the current version of Ang and Beniwal (2026).]

the dimensions of heterogeneity most relevant to predicting differential responses.

Pre-expansion Medicaid exposure is the most direct dimension. High-Medicaid-share hospitals stand to gain the most from expansion through increased billable volume and, in volume-responsive states, larger GME add-on payments. However, these institutions also carried the largest uninsured populations that generate DSH allotments, which are automatically reduced as uninsured volume falls. As Dranove, Garthwaite, and Ody (2016) show, the largest uncompensated care reductions post-expansion occurred at safety-net hospitals, meaning the net revenue effect depends critically on state DSH and GME formula design.

Hospital ownership structure affects investment behavior and responses to public policy. Duggan (2000) finds that private hospitals respond more aggressively than government-owned hospitals to Medicaid payment incentives, repositioning their patient mix in response to financial signals. In the training context, public and nonprofit hospitals may invest in residency capacity even without immediate financial returns, while for-profit institutions are more likely to condition expansion on realized revenue gains. Large academic medical centers with diversified revenue streams are better positioned to absorb above-cap training costs than community teaching hospitals on narrower margins.

Teaching intensity at baseline reflects the extent of fixed investments already made in residency infrastructure. Hospitals with higher pre-expansion resident-to-bed ratios face lower marginal costs of expansion and are more likely to respond to revenue improvements with additional slots. Hospitals with minimal training history face high entry costs that may prevent expansion even when revenues improve. Finally, urban hospitals benefit from deeper physician labor markets for faculty recruitment and attract more competitive applicant pools, while rural and micropolitan hospitals face structural constraints on both margins that limit their ability to translate financial gains into training capacity.

# 3 Conceptual Framework

It helps to fix ideas with a simple way of thinking about how residency positions are determined. Residency training has two sides: hospitals decide how many positions to create and fund, and medical graduates apply to fill them. The number of residents a hospital actually trains is the smaller of the two—the positions it offers, or the applicants willing to take them.

In practice, the offering side is what usually binds. In our data, the large majority of programs—82 percent overall, 80 percent in expansion states and 87 percent in never-expansion states—fill every position they offer (Table 1). For the typical program, then, the number of residents trained is set by how many positions the hospital is willing and able to fund, not by a shortage of applicants; the roughly one program in five with unfilled positions is a real margin, but a minority one. The question that matters is therefore what governs a hospital's decision to fund positions in the first place.

That decision is, above all, financial. Medicare pays for residents only up to a cap set in 1997; beyond that cap, a hospital must cover the full cost of each additional position—salary, benefits, and faculty supervision—out of its own operating budget. The marginal position is thus a discretionary expense that a hospital will sustain only if its finances allow, propped up by a mix of Medicare and Medicaid education payments and the disproportionate-share and uncompensated-care funds that help keep teaching hospitals afloat.

Medicaid expansion changes this financial picture, but not in a single direction. On one hand, it turns previously uninsured patients into paying—if low-margin—Medicaid patients, which improves revenue. On the other, it reduces the uncompensated-care payments that are tied to the size of the uninsured population, which cuts the other way. Whether a given hospital ends up with more or less room to fund positions depends on its patient mix and on how its state pays for graduate medical education (Section 2.1). The framework therefore delivers one clear prediction—that Medicaid expansion changes the number of residency positions by changing hospital finances—while leaving the *direction* of that change to be settled empirically (Section 6.3).

# 4 Data

We use data from the NRMP Main Residency Match reports for 2010 to 2019 (National Resident Matching Program 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019). These annual reports document, at the program level, positions offered, positions filled, match outcomes by graduate category, and fill rates, disaggregated by specialty. We identify the sponsoring institution for each program using the first four digits of the NRMP program code, which correspond to a unique hospital or training institution. Aggregating programs to the institution level yields a panel of 859 hospitals observed annually over ten years, for a total of 8,590 hospital-year observations when outcomes are summed across all specialties. Specialty-level analyses use the same panel structure but the number of observations varies because not all hospitals sponsor programs in every specialty. Because specialty-specific counts are conditional on a hospital sponsoring a program in that specialty, the pre-expansion means and per-program magnitudes reported by specialty and by hospital location need not sum to the all-specialty aggregate.

The primary outcome is matched positions per 100,000 population: the total number of positions filled at a given hospital in a given year, divided by the state population and scaled to a per-100,000 basis. Population data are from the U.S. Census Bureau. We analyze both the aggregate total across all specialties and specialty-specific counts. Heterogeneity by hospital type is examined in Section 6.4. Table 1 reports baseline (2010) summary statistics by expansion status: expansion and never-expansion states host programs of similar size, fill rates, rural composition, and specialty mix, though expansion states train substantially more physicians per capita—a level difference the program fixed effects absorb.

When the NRMP lists multiple specialties for a single program (e.g., "Emergency Medicine/ Anesthesiology"), slots are assigned to the first-listed specialty prior to aggregation to the hospital level. Because specialty models are estimated independently, this convention affects only the first-listed specialty's counts; the second specialty's count is unchanged. As a robustness check, we also allocate a slot to each listed specialty, which increases counts only for second-listed specialties.

Figure 1 summarizes the treatment structure of the panel. The unit of observation is a residency program (institution), and treatment is the host state's adoption of Medicaid expansion. States expanded in distinct waves—most prominently in 2014, with later adopters in 2015, 2016, and 2019—while a substantial set of hospitals reside in states that never expanded over the sample period and serve as the comparison group. This staggered rollout is the source of identifying variation for the event-study design described in Section 5.

Figures 2 and 3 plot the raw evolution of matched positions by expansion cohort, in totals and on a per-100,000 basis, where a cohort is defined by the calendar year in which a hospital's state expanded Medicaid. These series are purely descriptive and are not adjusted for fixed effects or covariates; they show that matched positions trend upward across all cohorts over the sample period, with treated and never-expansion hospitals moving broadly in parallel prior to expansion. The event-study estimates in Section 6 isolate the change in these trajectories attributable to expansion.

Finally, Figure 4 situates residency positions within the aggregate physician workforce. Using the American Community Survey, it plots the cumulative growth since 2001 in physicians per 100,000 population against growth in total population, each indexed to its 2001 value. Over nearly two decades physicians per 100,000 have risen at roughly the same pace as the population itself. Because the long-run supply of physicians is shaped upstream by the number of residency positions, this near-parallel growth motivates our focus on graduate medical education as the margin through which shocks to hospital finances may propagate into the physician workforce.

# 5 Empirical Strategy

In this paper, we estimate the dynamic effects of Medicaid expansion on the number of matched residency spots at a hospital per 100,000 state population using the imputation estimator developed by Borusyak, Jaravel, and Spiess (2024). This approach addresses the biases that arise when using conventional two-way fixed effects (TWFE) estimators in settings with staggered adoption difference-in-differences designs (De Chaisemartin and d'Haultfoeuille 2020, 2023; Goodman-Bacon 2021; Roth et al. 2023; Sun and Abraham 2021). We now discuss the model, identification assumptions, and estimation approach.

Let $Y_{ist}$ denote the number of matched residency spots per 100,000 state population at hospital $i$ in state $s$, and year $t$. Following Borusyak, Jaravel, and Spiess (2024), we specify the following event study model that allows for unrestricted treatment effect heterogeneity:

$$Y_{ist} = \lambda_i + \gamma_t + D_{ist}\tau_{ist} + \varepsilon_{ist} \quad (1)$$

where $\lambda_i$ represents hospital fixed effects, and $\gamma_t$ are time fixed effects. $D_{ist}$ is an indicator equal to one if Medicaid was expanded in state $s$ at time $t$, and $\tau_{ist}$ represents the fully heterogeneous treatment effect for hospital $i$ at time $t$. In equation (1), for each hospital, the data contains an activation date, $E_i$, when $D_{ist}$ switches from 0 to 1. This specification allows treatment effects to vary arbitrarily across hospitals and time periods without imposing parametric restrictions.

Our identification strategy leverages the staggered roll-out of Medicaid expansion across states between 2014 and 2019. The model in equation (1) is generated from three main assumptions on potential outcomes and causal effects. First, the parallel trends assumption requires that, absent Medicaid expansion, the number of matched residency spots per 100,000 state population would have evolved similarly between hospitals; because ours is an event-study design, we assess this assumption directly by testing for differential pre-expansion trends below. Second, we assume no anticipation—that Medicaid expansion did not affect the number of matched spots per 100,000 state population before the law's actual implementation in each state. This assumption is plausible in our setting: expansion timing was set by state-level political processes following *NFIB v. Sebelius* rather than by the training decisions of individual hospitals, and the multi-year lead times required to create or expand a residency program (Section 2.2.2) make preemptive adjustment unlikely.

A further assumption implicit in equation (1) is that potential outcomes are unaffected by other states' treatment—the stable unit treatment value assumption (SUTVA). It deserves explicit discussion here because residency positions are filled through a single national clearinghouse: if expansion leads treated-state programs to offer fewer positions, some applicants who would have matched there may instead match at programs in non-expansion states, mechanically raising outcomes in the comparison group. Spillovers of this form would inflate the treated–control gap, so our estimates should be read as an upper bound (in magnitude) on the per-state effect of expansion. Two features of the setting limit the scope for such interference. First, the relevant absorption margin is slack in the *comparison* group, where 87 percent of programs fill every position they offer (Table 1; the expansion-state rate is 80 percent), so comparison-state programs can absorb displaced applicants only where slack exists. Second, and more important, the number of positions a program *offers* (its quota) is set by hospital funding decisions before the match algorithm runs and cannot be inflated by the equilibrium reallocation of applicants. The offered-positions result in Section 6.1 is therefore the interference-robust object in our analysis: that quotas contract alongside matched positions indicates the effect reflects capacity decisions rather than an artifact of match-market reallocation across state lines.

Finally, we impose a model of unrestricted causal effects, referred to as the "null model" in Borusyak, Jaravel, and Spiess (2024). In this case, the target estimand (parameter of interest) is the dynamic average treatment effect on the treated (ATT) $h$ periods (horizons) since the treatment for a given $h \geqslant 0$:

$$\tau_h = \sum_{\{i,s,t\}:K_{ist}=h} w_{ist}\tau_{ist} \tag{2}$$

where $K_{ist} = t - E_i$ denotes event time—the number of years elapsed since Medicaid expansion in hospital $i$'s state—and the weight is given by $w_{ist} = \frac{\mathbb{1}(K_{ist}=h)}{|\{i,s,t\}:K_{ist}=h|}$ and sums to one within each event time $h$. Borusyak, Jaravel, and Spiess (2024) proposes an imputation estimator that uses untreated observations to predict what would have happened to treated units in the absence of treatment.

The estimator proceeds in three steps:

1. Using only the untreated units (i.e., observations with $D_{ist} = 0$) and ordinary least squares (OLS), we obtain $\hat{\lambda}_i$, $\hat{\gamma}_t$, and $\hat{\delta}$ from

$$Y_{ist} = \lambda_i + \gamma_t + X'_{ist}\delta + \varepsilon_{ist},$$

   where $X_{ist}$ is a vector of predetermined covariates. In our baseline specification $X_{ist}$ is empty—the counterfactual is built from the hospital and year fixed effects alone—and we use it only in robustness checks (e.g., adding baseline state population as a control).

2. For each treated observation $\{i, s, t\}$ with $D_{ist} = 1$, we construct the untreated potential outcome (counterfactual outcome) as $\hat{Y}_{ist}(0) = \hat{\lambda}_i + \hat{\gamma}_t + X'_{ist}\hat{\delta}$ and estimate the individual-specific treatment effect as $\hat{\tau}_{ist} = Y_{ist} - \hat{Y}_{ist}(0)$.

3. Estimate the event-time coefficients as weighted averages: $\hat{\tau}_h = \sum_{\{i,s,t\}:K_{ist}=h} w_{ist}\hat{\tau}_{ist}$.

We cluster standard errors at the state level to account for potential serial correlation within states over time that are robust to arbitrary forms of heteroskedasticity. Although the maintained assumptions of the difference-in-differences design are untestable in the post-treatment period, we can perform a robust test of the identifying assumptions in the pre-treatment period (pre-trends test). Unlike the conventional pre-trends test using standard event studies, the imputation-based method affords the opportunity to test for parallel pre-trends and no-anticipation assumptions using only the untreated observations.

To conduct the pre-trends test, one needs to choose an alternative model for the outcome $Y_{ist}$ for the untreated observations. Specifically, for an observable vector $W_{ist}$, the alternative model may be written as $Y_{ist} = \lambda_i + \gamma_t + X'_{ist}\delta + W_{ist}\theta + \varepsilon_{ist}$, where $W_{ist}$ may represent a set of binary indicators for $1, \ldots, k$ periods prior to the start of the treatment for some chosen $k$. Next, using the untreated observations only, obtain the OLS estimate of $\theta$ and test the hypothesis $\theta = 0$. We present all our main results graphically, combining these pre-trend estimates with the horizon-specific ATTs from equation (2). As discussed in Borusyak, Jaravel, and Spiess (2024), this robust OLS-based pre-trends test avoids the pre-testing concerns in Roth (2022). Specifically, regression-based tests use the full sample, including the treated observations, thereby imposing restrictions on treatment effect heterogeneity. Moreover, conducting inference using the imputation estimates of the ATT remains valid even if we condition on passing the pre-trends, avoiding the issue of inflated variances and overly conservative inference that often arises with standard pre-trend tests (Roth 2022).

# 6 Results

## 6.1 Main Effect of Medicaid Expansion on Residency Positions

Medicaid expansion is followed by a *reduction* in matched residency positions that emerges in the first post-expansion years and grows more negative over subsequent horizons. Figure 5 plots the dynamic treatment effects from the imputation estimator of Borusyak, Jaravel, and Spiess (2024), with the outcome measured as matched positions per 100,000 *contemporary* state population. The average post-expansion effect is $-0.016$ positions per 100,000, a decline of about 7.3 percent relative to the pre-expansion mean of 0.22 (joint $p = 0.10$). The pre-expansion coefficients are small and, if anything, slightly positive (joint pre-trend $p = 0.26$), consistent with parallel trends prior to expansion.

We scale by contemporary rather than fixed population deliberately. Normalizing matched positions by year-varying state population nets differential population growth out of the outcome and yields a specification with flat pre-trends; the raw *count* of positions per hospital, by contrast, exhibits a pronounced differential pre-trend (Appendix Figure A.2) that confounds a levels interpretation, so we treat levels as evidence on the direction of the effect only (see Section 7.3). Reassuringly, the contraction is if anything larger and more cleanly identified when we restrict identification to variation in the *timing* of expansion among adopters (the not-yet-treated design): the average effect there is $-0.033$ per 100,000, a 14.8 percent decline with flat pre-trends (joint $p < 0.01$; Appendix Figure A.17).

The reduction also appears at the stage of *offered* rather than filled positions: the quota (positions offered by programs prior to the matching algorithm) falls by 0.018 per 100,000, about 7.8 percent ($p = 0.03$). That the effect operates on offered positions indicates that the contraction reflects program capacity decisions rather than a change in how many offered slots go unfilled. As we detail in Section 7.3, the aggregate per-capita magnitude is estimated imprecisely—because expansion is assigned at the state level and identifying variation concentrates in a few large states, it is not statistically significant under randomization inference.

These balanced-panel estimates are, however, only the first half of the answer. The panel that generates them is exactly balanced (859 institutions $\times$ 10 years) because institution-years absent from the NRMP report were recorded as zeros, which treats the entry and exit of training institutions as changes in incumbent capacity. The next subsection shows that this margin carries most of the action, and that the honest reading of the evidence is reallocation rather than contraction.

## 6.2 The Extensive Margin: Entry, Exit, and Reallocation

Our data distinguish three objects the balanced panel conflates: intake at incumbent institutions, entry and exit of training institutions, and state-level training totals. Of the 859 NRMP sponsoring institutions, 223 first appear in the match after 2010—166 of them from 2016 onward, the window of the osteopathic (AOA-to-ACGME) single-accreditation transition—and 88 stop reporting positions before 2019. About a fifth of all institution-years (1,884 of 8,590) fall outside the window in which the institution appears in the NRMP report at all; the balanced panel records them as zeros.

Handling the margin explicitly changes the headline. Coding entry and exit years as missing rather than zero, the incumbent-margin effect is $-0.003$ per 100,000 ($-1.2$ percent) and statistically indistinguishable from zero; restricting the panel to the 560 institutions active in all ten years gives $-0.003$ ($-0.8$ percent), also indistinguishable from zero. (Both corrected panels exhibit differential pre-trends of their own—pre-trend $p < 0.001$—so we do not lean on their point estimates either; what is robust is that the balanced-panel contraction does not survive the correction.) Excluding only the single-accreditation-window entrants leaves a marginal contraction of $-6.7$ percent ($p = 0.05$), indicating that the entrant institutions are where the offsetting growth is concentrated: by 2019, post-2010 entrants account for roughly 2,200 matched positions in expansion states and 1,500 in never-expansion states. At the state level, where entry belongs in the total, per-capita training does not fall after expansion—the point estimate is positive (+2.9 percent; clustered $p = 0.05$, with a failing state-level pre-trend that again counsels caution about magnitude, not sign.)

The linked cost-report sample corroborates the incumbent-margin reading with an independent measure of the training stock. Matching NRMP institutions to Medicare provider numbers (700 of 859 institutions) and estimating match intake and annualized cost-report resident FTEs on the same rows, weights, and year convention, the two series move together: both decline modestly on the linked sample, and the apparent 19 percent FTE *rise* in volume-responsive states that emerges from the unweighted, unlinked cost-report panel does not appear on the comparable sample. Stock and flow tell one story once they are measured on the same institutions.

This decomposition also reconciles this paper with its companion. Ang and Beniwal (2026), working with state-level aggregates, find that expansion increased GME funding and expanded training capacity; the balanced institution panel here appeared to find the opposite. Both are correct about their own estimand: state aggregates include the entry margin, where capacity grew; the balanced panel weights incumbents, whose intake was flat to slightly negative. A policymaker asking "did expansion shrink my state's training pipeline?" should use the state-level object (no, and possibly the opposite); a hospital association asking "did incumbent programs grow?" should use the incumbent margin (no, roughly flat).

## 6.3 Mechanism: The Medicaid GME Financing Channel

The framework leaves the sign of the demand shift to the data because expansion changes hospital finances in offsetting ways (Section 2.1): it converts uncompensated care into billable Medicaid revenue and, in states whose Medicaid GME formulas scale with patient volume, raises GME add-on payments, while the Medicare uncompensated-care pool and below-cost Medicaid reimbursement push the other way. We isolate the operative channel using cross-state variation in Medicaid GME formula design (Section 2.2.3): *volume-responsive* states gain GME revenue when expansion raises Medicaid volume, whereas *fixed* and non-paying states gain none. States are classified from the payment methodology reported in the pre-period survey of Henderson (2013). If the financing channel operates, the contraction should be concentrated in the latter group. We present the evidence in the order we uncovered it: the 2012-vintage classification produces a striking cross-formula contrast, and the balance of this subsection lays it out. We then report the diagnostics that overturn it—reclassification to the post-period formula rules, leave-one-state-out estimation, and the linked cost-report sample—and conclude that the contrast is a California phenomenon rather than a mechanism. We keep the full presentation because the negative result is informative for the literature that uses these formula classifications.

As a direct test of the first link in this chain—whether expansion changes the GME revenue that funds training—Figures 6 and 7 estimate the effect of expansion on hospitals' GME payments using Centers for Medicare and Medicaid Services (CMS) Medicare cost-report data. Expansion raises Direct GME payments—which reimburse resident salaries, benefits, and supervision—by roughly 14 percent in the initial post-expansion years (joint $p < 0.01$; Figure 6), while the indirect medical education adjustment responds more weakly (Figure 7). Expansion thus demonstrably increases the training-specific resources flowing to teaching hospitals; the cross-formula split below shows that where this revenue gain is absent, residency positions contract.

Figure 8 reveals a sharp contrast consistent with the GME-financing channel. In non-responsive (fixed/none) expansion states, the average post-expansion effect is $-0.022$ per 100,000—an 11.7 percent decline relative to the pre-expansion mean of 0.18 (joint $p < 0.01$)—following flat pre-expansion coefficients (pre-trend $p = 0.15$). In volume-responsive states, where expansion mechanically raises Medicaid GME payments, the average post-expansion effect is $+0.013$ per 100,000 ($+3.4$ percent relative to a pre-expansion mean of 0.39): positive, with no decline at any horizon. The contraction in residency positions is thus concentrated in states whose Medicaid GME formulas do not reward higher Medicaid volume—exactly where the financing squeeze operates without an offsetting GME revenue gain. The contrast is statistically sharp under conventional inference: a pooled interacted model puts the volume-minus-non-responsive difference in average post effects at 0.035 per 100,000 (clustered $p < 0.001$). Under randomization inference—permuting expansion cohorts across states and recomputing the cross-arm difference each draw—the difference has $p = 0.19$: short of conventional thresholds, but notably more supportive than any individual estimate in the paper under that demanding standard.

Two caveats bound how much weight the volume-responsive arm of this comparison can bear. First, the volume-responsive panel exhibits a pronounced upward pre-trend (pre-trend $p < 0.001$), so its point estimate cannot be read as a clean counterfactual comparison; we treat it as descriptive. Second, the formula classification is a cross-state split, not a randomized one. Appendix Table A.2 documents baseline (2010) comparability across the two groups of expansion states: on state-level observables—positions per program, positions and quotas per 100,000, program counts, population, rural program share, and expansion timing—the groups are statistically indistinguishable (all $p > 0.17$). The population-*weighted* program-level baseline does differ (roughly 0.36 versus 0.17 matched positions per 100,000), reflecting the concentration of training in a few large volume-responsive states rather than state-level imbalance. Given both caveats, the mechanism claim rests on the two legs that survive them: the contraction in non-responsive states after flat pre-expansion coefficients, and the cross-formula contrast in the payment evidence below.

A related concern is correlated policy: the formula classification is a state policy choice, and other policies that moved hospital finances around expansion—exposure to the ACA's scheduled DSH reductions, hospital provider taxes, Medicaid managed-care penetration, overall Medicaid payment generosity—could in principle differ across the two groups and generate their own cross-arm contrast. Two features of the evidence limit this concern. First, the classification is measured in the 2012 pre-period, the groups are balanced on baseline observables, and their expansion timing is identical (Appendix Table A.2), so a confounding policy would have to operate through channels invisible in the baseline comparison. Second, a confounder would need to reproduce a specific fingerprint rather than generic fiscal relief: volume-responsive states show no slot decline, a 19 percent rise in cost-report resident FTEs, and an IME payment response, while non-responsive states show slot declines with flat FTEs—the pattern a volume-scaled training subsidy produces and that broad financial improvement does not. (As shown below, the FTE half of this fingerprint does not survive estimation on the linked sample.)

The financing channel makes a sharper prediction about the payment response itself: if the mechanism operates, the GME revenue gain from expansion should be concentrated in volume-responsive states and largely absent in non-responsive ones. Appendix Figures A.13 and A.14 assess this by re-estimating the payment event study separately by formula design. Two measurement caveats apply. The payments observable at the hospital level are *Medicare* cost-report DGME and IME amounts—state Medicaid GME payments are not reported at the hospital level—so we read them as a proxy for training-related revenue rather than a direct measure of the Medicaid GME channel. And because realized Medicare payments move with resident counts and patient-share terms, part of any post-expansion rise is common to all expansion states; the mechanism-relevant object is the cross-formula *difference*, not the level—and we test each difference directly. For Indirect Medical Education payments the contrast is sharp: payments rise by roughly 17 percent in volume-responsive states (joint $p < 0.01$) and not at all in non-responsive states ($-0.3$ percent, $p = 0.30$), a cross-group difference with $p = 0.001$. For Direct GME, the levels contrast (24 versus 10 percent, each $p < 0.01$) is large but imprecise as a difference ($p = 0.15$). Decomposing Direct GME into resident FTEs and payment per FTE (Appendix Figures A.15 and A.16) is more informative. In volume-responsive states the payment growth is a *quantity* response: cost-report resident FTEs rise by roughly 19 percent (joint $p < 0.001$) while payment per FTE does not rise. In non-responsive states FTEs do not move at all ($+3$ percent, $p = 0.95$; cross-group FTE difference $p = 0.005$), so the residual Direct GME rise there occurs *without* any growth in residents—consistent with the mechanical rate and patient-share channels just described rather than with a training expansion. The decomposition also resolves an apparent tension: Direct GME can rise in non-responsive states even as match intake falls precisely because the payment growth there is not a resident-count phenomenon. On the 2012 classification and the unlinked cost-report panel, then, the training-quantity response appears concentrated in the states where residency positions do not contract.

### Reclassification, leave-one-out, and the linked sample overturn the mechanism

Three diagnostics, each requested by the logic of the design itself, undo this reading. First, *reclassification*. Our formula classification file records a second vintage reflecting 2015 payment rules, under which New York, Illinois, Arizona, Vermont, and Virginia move *into* the volume-responsive arm (and North Carolina and Idaho, both non-expansion states, move out). Because the treatment window is 2014–2019, the 2015 vintage—not the 2012 one—describes the rules in force for essentially the entire post period; an exposure-weighted time-varying classification coincides with it for every state. Under the 2015 vintage the cross-arm difference falls from 0.035 to 0.024 per 100,000 (delta-method $p < 0.001$; robust to flipping each of the five states whose coding required judgment), the volume arm's failing pre-trend is cured ($p = 0.39$), but the failing pre-trend reappears in the non-responsive arm ($p < 0.001$)—the instability travels with the reclassified states rather than disappearing. More decisively, the payment-side contrast vanishes: under the 2015 vintage DGME payments rise in *both* arms ($+17$ and $+11$ percent) and the cross-arm difference has $p = 0.45$.

Second, *leave-one-state-out*. Dropping California—coded non-responsive because Medi-Cal pays no fee-for-service GME, a coding we document and defend in the appendix, but a coding on which everything turns—flips the sign of the 2015-vintage cross-arm difference (from $+0.024$ to $-0.009$, not significant). The same happens inside the not-yet-treated design (+0.021 full sample, −0.008 without California). No other state's removal moves the contrast appreciably. Under wild cluster bootstrap-t inference the 2015-vintage difference has $p = 0.39$ (Section 7.3).

Third, the *linked sample*. Estimating cost-report resident FTEs and match intake on the same institutions with the same weights and year convention, FTEs do not rise 19 percent in volume-responsive states; they decline modestly in both arms, tracking intake. The FTE fingerprint that anchored the mechanism's corroborating evidence is an artifact of comparing a population-weighted NRMP panel with an unweighted, differently-sampled cost-report panel.

We conclude that the data do not establish a Medicaid GME financing mechanism. What they establish is narrower and still useful: the cross-formula contrast in published state classifications is dominated by California, and any future work splitting states on these classifications should report the reclassified and leave-one-out versions of its results.

## 6.4 Heterogeneous Effects by Hospital Location

The aggregate effect masks heterogeneity across hospital settings. Figure 9 estimates the effect separately for urban and rural hospitals in two *split-sample* regressions: each group is compared only with never-expansion programs of the same type, and—the substantive gain over a pooled interacted model—each group carries its own pre-trends test. For urban hospitals, which account for the large majority of training volume, the average post-expansion effect is −0.016 per 100,000 (joint $p = 0.10$), a 6.6 percent decline. The urban pre-expansion coefficients are uniformly *positive* (joint $p = 0.04$): urban programs in expansion states were on a slightly rising differential path before expansion, so the post-expansion decline is a reversal in sign rather than the continuation of a prior trend. The direction of any bias works against the finding: a rising differential pre-path, if it continued, would push the imputed counterfactual up and the estimated effect toward zero, so the urban decline is, if anything, conservative. A Rambachan and Roth (2023) sensitivity analysis for the urban subsample is reported alongside the headline in Appendix Figure A.20.

For rural hospitals, the split-sample design materially changes the reading relative to a pooled specification. The average post-expansion effect is −0.020 per 100,000 (joint $p < 0.001$), which against a rural baseline of only 0.03 positions per 100,000 would amount to more than half of baseline rural training—but the rural subsample *fails its own pre-trends test* decisively (joint $p < 0.001$), with pre-expansion coefficients of −0.02 to −0.04 per 100,000, comparable in size to the post-expansion effects themselves. Rural programs in expansion states were already losing ground relative to rural programs in non-expansion states before expansion. A pooled interacted model, which by construction shares a single set of pre-trends across the two groups, puts the urban–rural difference in average post effects at −0.014 per 100,000 ($p = 0.02$); but because that model pools exactly the pre-trends the split samples show to be different, we treat the rural magnitude and the difference test as descriptive rather than causal. The formal multiple-testing evidence points the same way: under clustered inference the rural effect is the lone survivor of the false-discovery-rate correction across our family of tests ($q = 0.002$), but it does not survive when the correction is built on randomization-inference p-values ($p = 0.29$; Appendix Table A.1), and the failed pre-trends caution against reading the clustered survival causally. What the location split supports is the narrower statement that the contraction is not confined to rural margins: it appears at the urban programs where most training occurs, and per-capita training at rural hospitals—where graduate medical education capacity is already thin—was falling in expansion states throughout the period.

The rural–urban gap is qualitatively similar under the fixed-2010 per-capita and levels specifications, estimated with the same split-sample design (Appendix Figures A.4 and A.5), though the levels magnitudes are inflated by the differential pre-trend in the raw count discussed in Section 7.3.

### 6.5 Heterogeneous Effects by Specialty

We next examine whether the effect differs across specialties, distinguishing primary care (family medicine, internal medicine, and pediatrics) from all other specialties. Figure 10 presents the estimates. Both groups show a proportional contraction of similar size: the average post-expansion effect is $-0.003$ per 100,000 for primary care, a 4.6 percent decline ($p = 0.06$), and $-0.002$ per 100,000 for non-primary care (4.7 percent; $p = 0.22$). Neither subgroup effect is precisely estimated on the year-varying outcome, and both panels exhibit differential pre-trends—the non-primary-care panel fails the pre-trends test ($p = 0.01$) and the primary-care panel is marginal ($p = 0.06$)—so we read the specialty split as suggestive rather than definitive.

Under the fixed-2010 per-capita and levels specifications the point estimates are proportionally larger for primary care (Appendix Figures A.6 and A.7), but those specifications share the differential-pre-trend and imprecision caveats discussed in Section 7.3. Because primary care is the principal pipeline into the physician specialties that face the most acute shortages, any contraction concentrated there would be of particular concern; establishing it more sharply is a task for richer data than a state-level design with few clusters can support.

## 7 Robustness Checks

### 7.1 Alternative Difference-in-Differences Estimators

To demonstrate that our findings are not an artifact of the particular estimator we adopt, we re-estimate the event study using a suite of recently developed difference-in-differences estimators that are robust to treatment-effect heterogeneity under staggered adoption.

Figure 11 compares six approaches for matched positions per 100,000 contemporary population: two-way fixed effects (TWFE), Borusyak, Jaravel, and Spiess (2024), De Chaisemartin and d'Haultfoeuille (2020), Sun and Abraham (2021), Cengiz et al. (2019), and Gardner (2022). Each addresses the "forbidden comparisons" that can contaminate conventional TWFE estimates under staggered timing through a different methodological route. Across estimators, the post-expansion estimates align closely in both sign and magnitude, and the pre-expansion coefficients are near zero throughout.[3] The same robustness holds for the levels outcome (Appendix Figure A.8) and the quota outcome (Appendix Figure A.9).

## 7.2 Unweighted Specifications

Our baseline specifications weight observations by state population so that the estimates summarize the experience of the average resident rather than the average program. As a robustness check, Appendix Figures A.10 to A.12 re-estimate the main results without population weights. The unweighted estimates preserve the sign and dynamic shape of the baseline results—a negative, widening post-expansion effect with flat pre-trends—though they place relatively more weight on small programs and are less precise in the per-capita specifications. The unweighted per-capita point estimates are not attenuated: in absolute value they are similar to or larger than the weighted ones (for the year-varying denominator, −0.017 unweighted against −0.016 weighted), but they sit on a much larger baseline mean (0.50 against 0.22), so the implied percentage effect is smaller (−3.4 against −7.3 percent) and the standard error is wider. In levels, the unweighted effect on matched positions remains statistically significant (−2.49 positions per hospital, −8.5 percent; $p < 0.01$; Appendix Figure A.10); the per-capita matched and quota effects retain the same sign but are not statistically significant (Appendix Figures A.11 and A.12; the unweighted estimate on the headline's own year-varying denominator, completing the weighting-by-denominator grid, behaves the same way). The contrast between the weighted and unweighted per-capita estimates indicates that the statistical significance of the per-capita headline depends on population weighting: the precision is driven by the large programs that account for most residency training. We therefore read the per-capita estimates as evidence on the direction and locus of the effect rather than as a precise population-invariant magnitude, and we place correspondingly more weight on the levels and offered-position results—which are significant with and without weights—and on the mechanism evidence.

[3] We exclude the Callaway and Sant'Anna (2021) estimator from this comparison because it is not well identified given the sparsely populated never-treated cells in our sample. Differences across the remaining estimators can arise from the choice of weights used to aggregate group-time effects into an overall ATT; see, e.g., Deb et al. (2025).

## 7.3 Inference with Few Clusters and Sensitivity to Parallel Trends

Because Medicaid expansion is assigned at the state level and the population-weighted design concentrates identifying variation in a small number of large expansion states, the effective number of clusters is well below the fifty-one states used to compute the clustered standard errors, and the asymptotic p-values should be read with caution. The concentration is quantifiable: following the logic of Carter, Schnepel, and Steigerwald (2017) and approximating the effective number of clusters by $G^* = G/(1 + CV^2)$, where CV is the coefficient of variation of state shares of the estimation weight, the weighted design has $G^* \approx 7$ (the five largest states carry 68 percent of the weight), against $G^* \approx 21$ without weights. We subject the main results to several checks, and we are candid that they place the aggregate magnitude on softer statistical ground than its sign and mechanism.

First, randomization inference that permutes the vector of expansion-timing cohorts across the states in each specification's sample under the sharp null of no effect—re-estimating the average post-expansion effect over 1,000 permutation draws for the headline and mechanism specifications and 500 draws for the remaining family members—yields conservative p-values: about 0.58 for the headline effect, 0.32 for the non-responsive mechanism panel, and 0.75 for the volume-responsive panel. Applied uniformly to the other estimates the paper leans on, the same exercise gives 0.38 for the not-yet-treated design, 0.29 for the rural subsample, 0.63 for urban, 0.51 for offered positions, 0.76 and 0.48 for primary and non-primary care, and 0.19 for the volume-minus-non-responsive mechanism difference. Under this standard, no individual estimate can be distinguished from zero, though the mechanism difference comes closest. Because the permutation distribution of the population-weighted estimator is dominated by which large states are assigned to treatment, this test is demanding in exactly the way the design invites; we read it as indicating that none of the individual magnitudes is estimated precisely enough to reject zero under the most conservative benchmark—not that the pattern of results is uninformative.

Second, a Rambachan and Roth (2023) relative-magnitudes sensitivity analysis (Appendix Figure A.20) is consistent with this reading: the 95 percent robust confidence interval for the average post-expansion effect includes zero even under exact parallel trends. What survives is the *pattern*—a negative, monotonically growing dynamic path—and the joint tests of the post-expansion coefficients, which reject under clustered inference (though not under randomization inference) for the non-responsive mechanism panel and for the not-yet-treated design.

Third, the estimates that are sharpest under conventional clustered inference—the not-yet-treated design and the rural subsample—illustrate why we report both inference standards rather than choosing one. Restricting the sample to eventually-treated states, so that identification relies solely on differences in the timing of expansion, yields an average per-capita effect of $-0.033$ (14.8 percent; joint clustered $p < 0.01$; Appendix Figure A.17) with flat pre-trends, and the rural effect is the only member of the primary family that survives a Benjamini–Hochberg false-discovery-rate correction built on the clustered p-values ($q = 0.002$; the full-sample, urban, primary-care, and offered-position effects have clustered q-values between roughly 0.10 and 0.13; Appendix Table A.1 and Figure A.1). But neither result survives the conservative standard: their randomization-inference p-values are 0.38 and 0.29, and when the false-discovery-rate correction is rebuilt on the randomization p-values, no member of the family survives ($q \geqslant 0.76$). The rural estimate additionally fails its own pre-trends test (Section 6.4). We therefore claim statistical significance for no individual subgroup under uniform conservative inference; the evidentiary weight rests on the sign and growing dynamic path common to every specification, the offered-positions margin (which is robust to match-market interference), and the cross-formula mechanism contrast.

Fourth, two further diagnostics probe the design's declared weak points. Because horizons +4 and +5 are identified only from early adopters, the growing dynamic path could in principle reflect cohort composition rather than dynamics. Restricting treatment to the 2014 cohort alone (Appendix Figure A.18) holds composition fixed and reproduces both the magnitude (−0.015 per 100,000, a 7.5 percent decline; joint $p = 0.16$ in the smaller single-cohort sample) and the monotonically widening path—the event-time coefficients grow from −0.004 at horizon 0 to −0.024 at horizon +5—after pre-expansion coefficients that are as flat as the design can hope for (pre-trend $p = 0.97$). And leave-one-state-out estimation (Appendix Figure A.19; the figure reports 95 percent confidence intervals, and the full-sample benchmark is estimated on the same automatically-trimmed sample as the leave-one-out runs) makes the concentration of identifying variation concrete: every one of the 34 leave-one-out estimates over treated states remains negative (range −0.018 to −0.001), but dropping California alone shrinks the average effect from −0.016 to −0.001. Extending the exercise to control states shows the comparison group matters too: dropping Texas—a never-expansion state that ran large state GME appropriations over the same window—moves the estimate to −0.032, twice the benchmark, and excluding the four control states with the largest competing GME programs (Texas, Florida, Georgia, Tennessee) yields −0.042 (−19 percent; joint $p = 0.02$). The sign of the effect does not depend on any single state; the magnitude leans heavily on the largest treated state and is, if anything, held down by GME-expanding control states—consistent with the effective-cluster calculation above and with our decision to emphasize the pattern of the effect over its point estimate.

Fifth, two inference diagnostics that earlier drafts left unreported belong on the page. A randomization-inference diagnostic that varies the outcome and the weighting shows that the *levels* specifications are significant under the same conservative standard that the per-capita family fails: permuting expansion timing yields $p = 0.010$ for weighted levels and $p = 0.008$ for unweighted levels, against $p = 0.51$ for the unweighted fixed-denominator per-capita outcome. We read this asymmetry as informative rather than reassuring: the levels outcome carries the differential pre-trend documented above, so its randomization significance partly reflects that trend, and the per-capita outcome that removes the trend also removes the significance. Separately, wild cluster bootstrap-t inference (Webb weights, state clusters, 9,999 replications) on the static two-way fixed-effects analogs of our estimates—the bootstrap cannot be combined with the imputation estimator directly—gives $p = 0.41$ for the headline, $p = 0.50$ for the not-yet-treated design, and $p = 0.05$ for the cross-formula mechanism difference under the baseline classification ($p = 0.39$ under the 2015 classification vintage discussed in Section 6.3). These results reinforce the paper's stated inferential posture: the aggregate magnitude is not distinguishable from zero under conservative standards.

Finally, a word on the choice of outcome. We scale by contemporary population because the raw *count* of positions fails the parallel-trends test badly: its pre-expansion coefficients are large and negative—on the order of −2 positions per hospital, comparable to the treatment effect itself (Appendix Figure A.2)—so a levels estimate, though precisely estimated and negative, would largely reflect a pre-existing differential trend rather than a causal effect. We therefore report levels only as corroborating the *direction* of the effect, and we place the weight of our quantitative claims on the year-varying per-capita specification, whose pre-trends are flat. We acknowledge the pre-testing concern this ordering creates (Roth 2022): an outcome adopted in part because it passes a specification test will, by construction, pass that test in sample. Three considerations argue the choice is substantive rather than opportunistic. First, the per-capita rate is the natural policy object—training capacity relative to the population it serves—and scaling by contemporary population removes differential population growth, a component of the levels series that is mechanical rather than causal. Second, the imputation-based pre-trends test uses only untreated observations, so inference conditional on passing it remains valid (Section 5). Third, the conclusions we lean on do not depend on the pivot: the offered-positions result and the cross-formula mechanism contrast are similar under the fixed-2010 per-capita construction specified before the year-varying outcome was adopted, and the sign and dynamics of the contraction appear in every outcome definition we examine.

# 8 Policy Implications

What is at stake, in positions and dollars, is bounded by the estimates rather than asserted. At the incumbent margin, the corrected effect of −1.2 percent carries a 95 percent confidence interval that comfortably includes zero and, at its most negative edge, magnitudes on the order of the balanced-panel estimate; at the state level, where entry belongs in the total, the point estimate is mildly positive. Converting the most adverse plausible reading to dollars for perspective: public GME support runs roughly \$150,000 to \$180,000 per resident-year (Medicare DGME and IME totaled about \$16.2 billion; (Congressional Research Service 2022; Eden, Berwick, and Wilensky 2014)), so even a persistent loss of one thousand annual positions—well outside our state-level estimates—would represent roughly \$0.5–0.6 billion per year of forgone public training investment, against Medicaid GME spending of \$7.39 billion in 2022 (Henderson 2023). Because 50 to 60 percent of residents practice in their training state (Murphy 2020; Orr 2023), reallocation across institutions *within* states matters less for state physician supply than contraction would have; the incidence question our data can address is institutional (which sponsors train) rather than geographic.

Three implications follow. First, and most important to state plainly: nothing in this paper supports declining to expand Medicaid to protect the training pipeline. The corrected estimates show no aggregate contraction, the coverage literature documents large health and financial benefits of expansion (Miller, Johnson, and Wherry 2021; Sommers et al. 2017), and seven of the ten remaining non-expansion states have volume-responsive GME formulas—the configuration in which even the fragile cross-formula reading would predict no training loss. Second, the policy lever our results do speak to is panel-entry infrastructure: the capacity growth we document arrived through newly accredited sponsoring institutions, many via the single-accreditation transition, which suggests that accreditation pathways and start-up support for new programs—the margin targeted by Teaching Health Center GME, Section 5503 redistribution, and the CAA 2021 Section 126 slots—move training capacity in ways that incumbent-hospital financing formulas evidently did not. Third, for states weighing volume-responsive Medicaid GME formulas, the federal match makes conversion a comparatively cheap lever, but our evidence no longer supplies a demonstrated training return; the honest statement is that the formula margin is untested once California is set aside.

# 9 Conclusion

How many physicians the United States trains—and in which specialties and places—is settled years upstream, in the residency positions that teaching hospitals choose to create and fund. Those decisions are, at bottom, financial, which leaves the training pipeline exposed to any policy that moves hospital revenue. This paper asks whether the ACA's Medicaid expansion, by reshaping the finances of teaching hospitals, altered the number of residency positions those hospitals offer and fill. Using the staggered rollout of expansion across states and the imputation estimator of Borusyak, Jaravel, and Spiess (2024), we find that the answer is reallocation rather than contraction. A balanced institution panel shows an apparent decline in matched positions, but the decline does not survive honest treatment of the extensive margin: intake at incumbent institutions is roughly flat once entry and exit are coded as missing rather than zero, training capacity grows through institutions entering the match—many during the osteopathic single-accreditation transition—and state-level totals do not fall. The offered-position margin behaves like the matched margin throughout, and the linked cost-report sample shows resident stocks tracking intake on comparable institutions. This decomposition reconciles the hospital-level and state-level evidence, including the companion estimates in Ang and Beniwal (2026).

We also report, as a negative result, the fate of the financing mechanism earlier drafts advanced. Under the 2012 formula classification the contraction appears concentrated in states whose Medicaid GME formulas do not reward patient volume; under the post-period 2015 classification the payment-side contrast disappears, and the reduced-form contrast reverses sign when California leaves the sample. The cross-formula comparison in published classifications is, in these data, a California phenomenon.

Two features of the setting temper how precisely any of these magnitudes can be pinned down. First, treatment varies at the state level and the population-weighted design draws its identifying variation from a small number of large expansion states, so the effective number of clusters is limited; under randomization inference and wild cluster bootstrap, no individual estimate in the paper—incumbent margin, state total, or cross-formula difference—is statistically distinguishable from zero, and we say so wherever the estimate appears. Second, every panel construction carries its own pre-trend liability: the balanced panel is flat but zero-filled, the corrected panels are honest but fail pre-trends, and the contemporary-population denominator that delivers flat pre-trends is itself trending differentially (we show this directly by using log population as the outcome). Our confidence in the reallocation reading rests on its *consistency across constructions with offsetting weaknesses*, not on any single clean design. Finally, our data capture positions offered and filled, not the downstream practice locations or health outcomes of the physicians those positions would have trained; linking training-capacity changes to eventual physician supply and population health is an important direction we leave to future work.

Read together, the results suggest that the physician-training pipeline absorbed the largest coverage expansion in a generation without contracting—but that it did so through entry and reallocation rather than through incumbent growth, and that the reallocation is invisible to balanced hospital panels and state aggregates alike unless the extensive margin is measured deliberately. Which institutions train the next generation of physicians changed more than how many were trained. That compositional margin—who enters training provision, where, and under which accreditation and financing rules—is a natural target for future research and, we would argue, the policy-relevant one.

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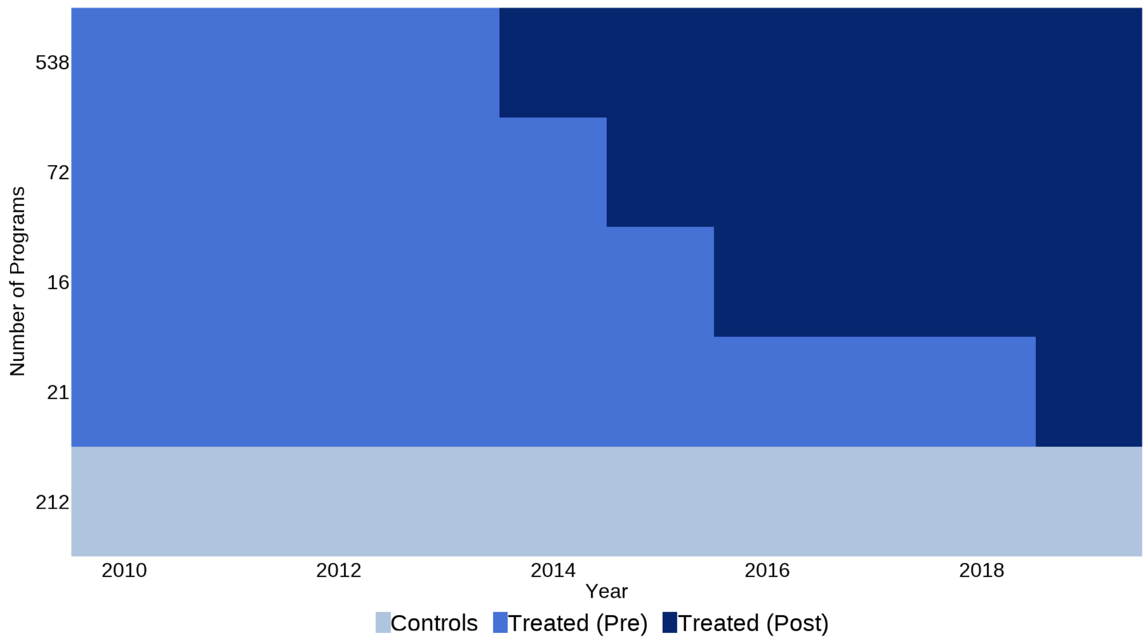
Variation Across States." *Health Affairs* 40 (2): 343–348. https://doi.org/10.1377/hlthaff.2020.00611.

Table 1: Summary Statistics at Baseline (2010)

| Baseline characteristic (2010) | All states (1) | Expansion states (2) | Never-expansion (3) | Difference (2)−(3) | p-value |
|---|---|---|---|---|---|
| Matched positions per program | 28.7 | 28.0 | 30.0 | -2.1 | 0.704 |
| Offered positions (quota) per program | 30.2 | 29.5 | 31.6 | -2.0 | 0.719 |
| Share of programs filling every offered position | 0.82 | 0.80 | 0.87 | -0.07 | 0.130 |
| Matched positions per 100,000 population | 7.95 | 9.19 | 5.46 | 3.74 | 0.046 |
| Offered positions (quota) per 100,000 | 8.36 | 9.66 | 5.76 | 3.90 | 0.042 |
| Number of programs | 16.8 | 19.0 | 12.5 | 6.6 | 0.197 |
| State population (millions) | 6.1 | 5.9 | 6.3 | -0.4 | 0.834 |
| Rural program share | 0.09 | 0.09 | 0.10 | -0.01 | 0.795 |
| Primary-care share of matched positions | 0.55 | 0.54 | 0.57 | -0.03 | 0.554 |
| Mean expansion year | 2014.5 | 2014.5 | – | – | – |
| Number of states | 51 | 34 | 17 | | |
| Number of programs | 859 | | | | |
| Hospital-year observations (2010–2019) | 8,590 | | | | |

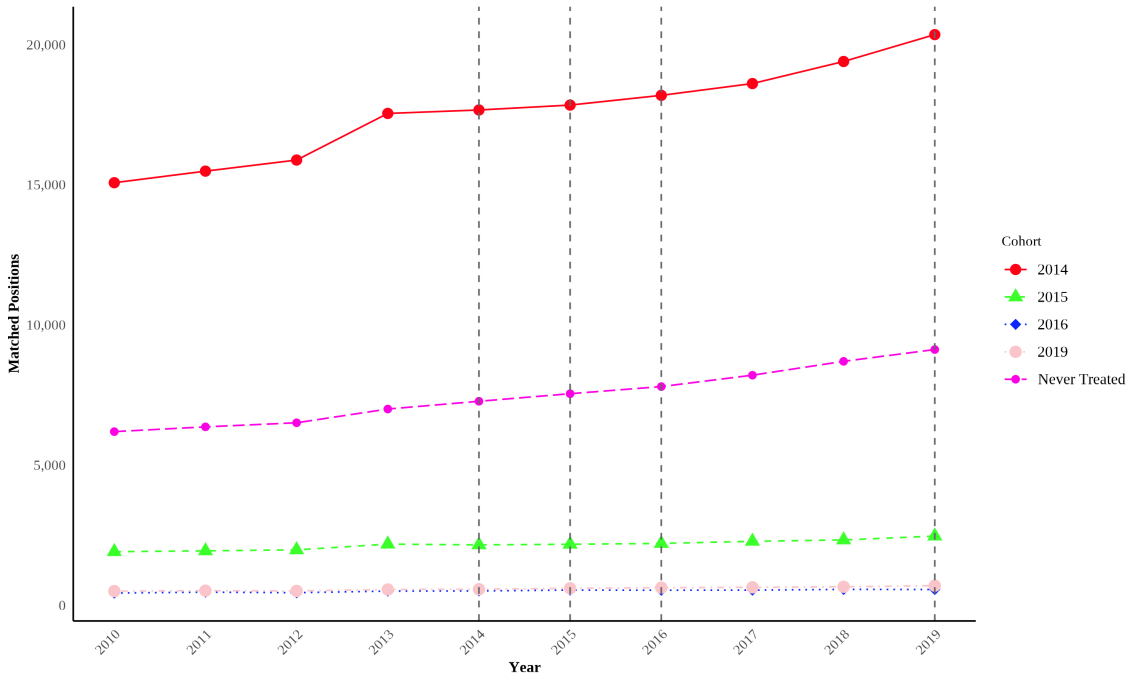
*Notes:* The table reports baseline (2010) means with the state as the unit of observation: each entry is the unweighted mean across states of the state-level value. Column (1) covers all 50 states and the District of Columbia; columns (2) and (3) split states by whether they ever adopted the ACA Medicaid expansion during 2010–2019; the p-value is from a Welch two-sample t-test of the expansion-versus-never difference. Program-level rows (matched and offered positions per program, rural share) first average across the state's programs, where a program is a sponsoring institution aggregated across specialties. The fill-share row is measured at the NRMP specialty-program level—the share of a state's specialty programs that fill every position they offer—matching the concept referenced in Section 3. Rural programs are those in counties with 2010 RUCA codes above 3; the primary-care share is the share of matched positions in family medicine, internal medicine, and pediatrics. Expansion states train more physicians per capita at baseline; the estimation design absorbs such level differences through program fixed effects. Data are from NRMP Main Residency Match reports and the U.S. Census Bureau.

Figure 1: Staggered Timing of Medicaid Expansion Across Residency Programs



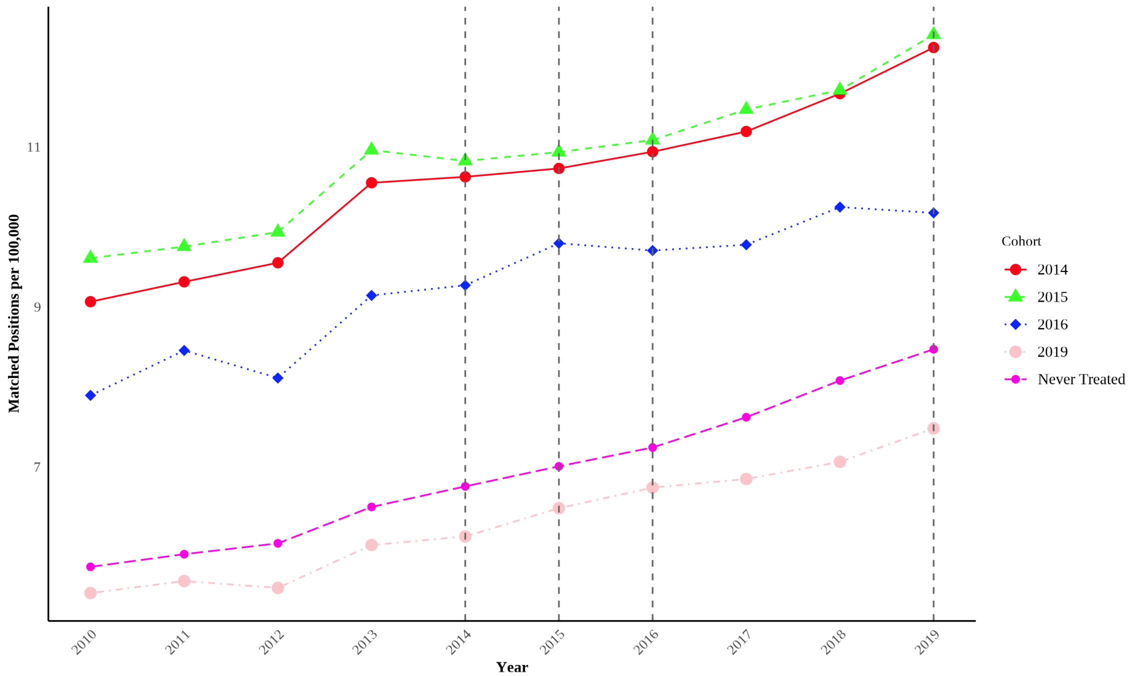

*Notes:* The figure summarizes the treatment structure of the sample, where the unit of observation is a residency program (institution) and the treatment is the host state's adoption of Medicaid expansion. Programs are grouped by the timing of their state's expansion, and the vertical axis reports the number of programs in each group. Medium-blue cells denote treated programs in years before their state expanded Medicaid ("Treated (Pre)"), dark-navy cells denote treated programs in expansion and post-expansion years ("Treated (Post)"), and light-blue cells denote never-expansion (control) programs. The horizontal axis is the calendar year. The figure illustrates the staggered rollout of Medicaid expansion exploited by the difference-in-differences design. Data are from NRMP Main Residency Match reports, 2010–2019. The plot is produced with the `panelview` package (Mou, Liu, and Xu).

Figure 2: Total Matched Residency Positions by Expansion Cohort




*Notes:* The figure plots the total number of matched residency positions by year, separately for each Medicaid-expansion cohort. A cohort is defined by the calendar year in which a program's state expanded Medicaid (2014, 2015, 2016, and 2019); programs in states that never expanded over the sample period are grouped as "Never Treated." Each series sums matched positions across all programs in the cohort, where positions are first aggregated to the institution-year level across specialties. Dashed vertical lines mark the expansion years of the treated cohorts. The figure is descriptive and is not adjusted for fixed effects or covariates. Data are from NRMP Main Residency Match reports, 2010–2019.

Figure 3: Matched Residency Positions per 100,000 Population by Expansion Cohort

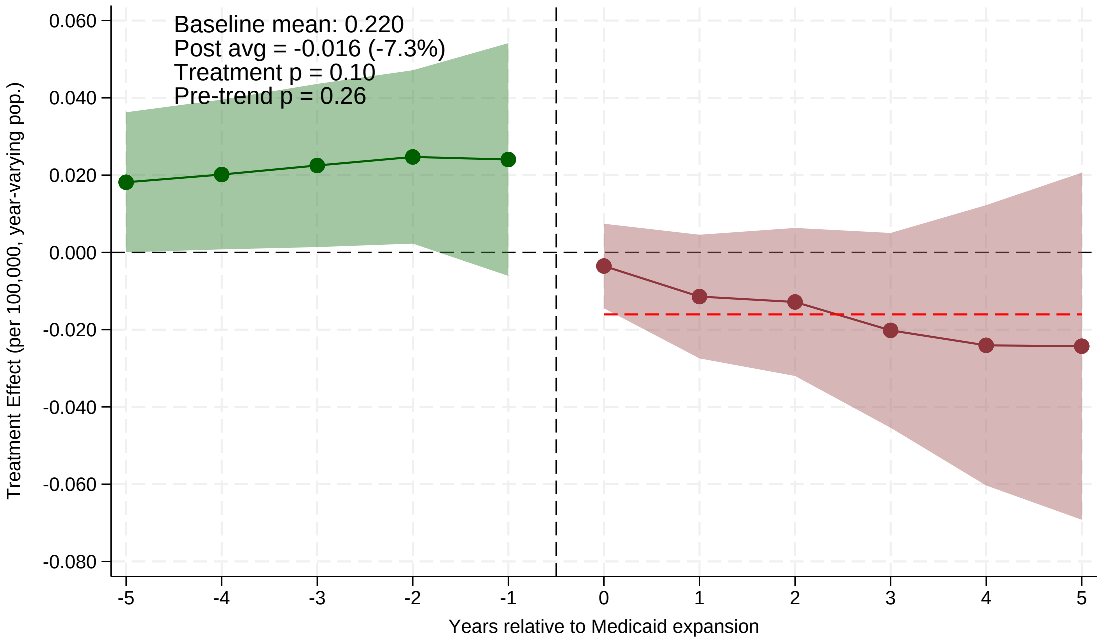



*Notes:* The figure plots matched residency positions per 100,000 population by year, separately for each Medicaid-expansion cohort. Cohorts are defined by the calendar year in which a program's state expanded Medicaid (2014, 2015, 2016, and 2019), with programs in non-expansion states grouped as "Never Treated." For each cohort and year, matched positions and population are summed across the cohort's states before computing the rate per 100,000, using 2010 state population. Dashed vertical lines mark the expansion years of the treated cohorts. The figure is descriptive and is not adjusted for fixed effects or covariates. Data are from NRMP Main Residency Match reports, 2010–2019, and U.S. Census Bureau state population estimates.

Figure 4: Growth in Physicians per 100,000 Population Relative to Total Population

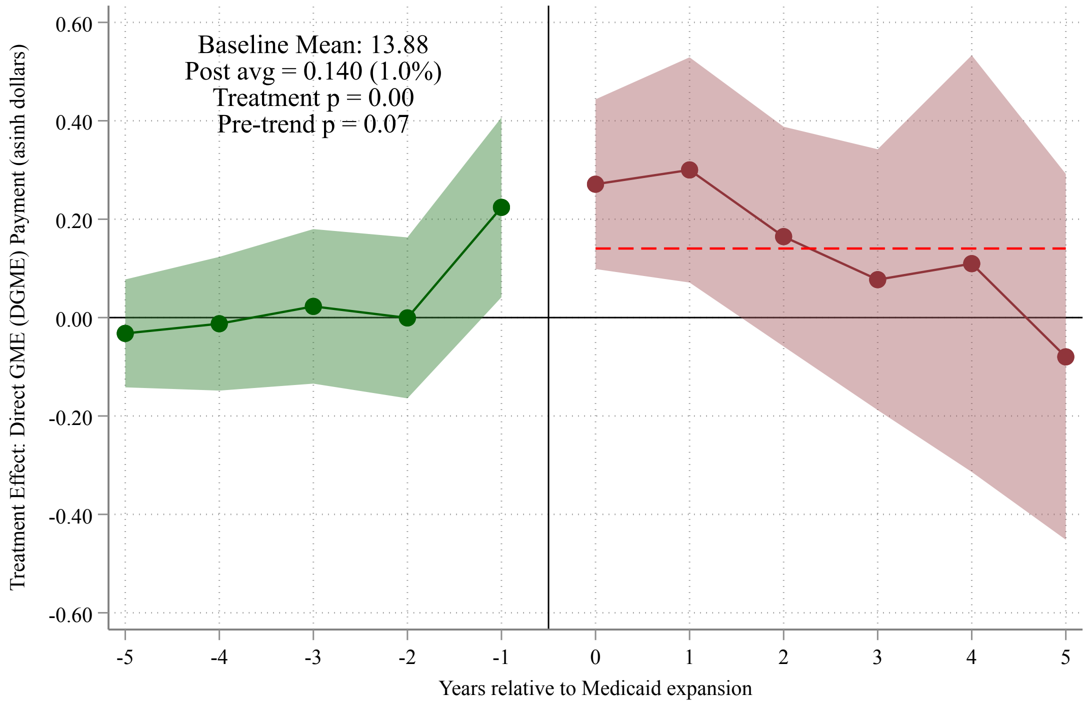

*Notes:* The figure plots the cumulative percent change since 2001 (2001 = 100) in two national series from the American Community Survey (ACS): physicians per 100,000 population and total population. Physicians are defined as employed workers in the harmonized occupation code for physicians and surgeons (`OCC1950` = 75). Estimates use the ACS person weight (`PERWT`). Data are from IPUMS-USA. The figure motivates the analysis by showing that the physician-to-population ratio has roughly tracked, rather than outpaced, population growth.

Figure 5: Effect of Medicaid Expansion on Matched Residency Positions: Event Study Estimates



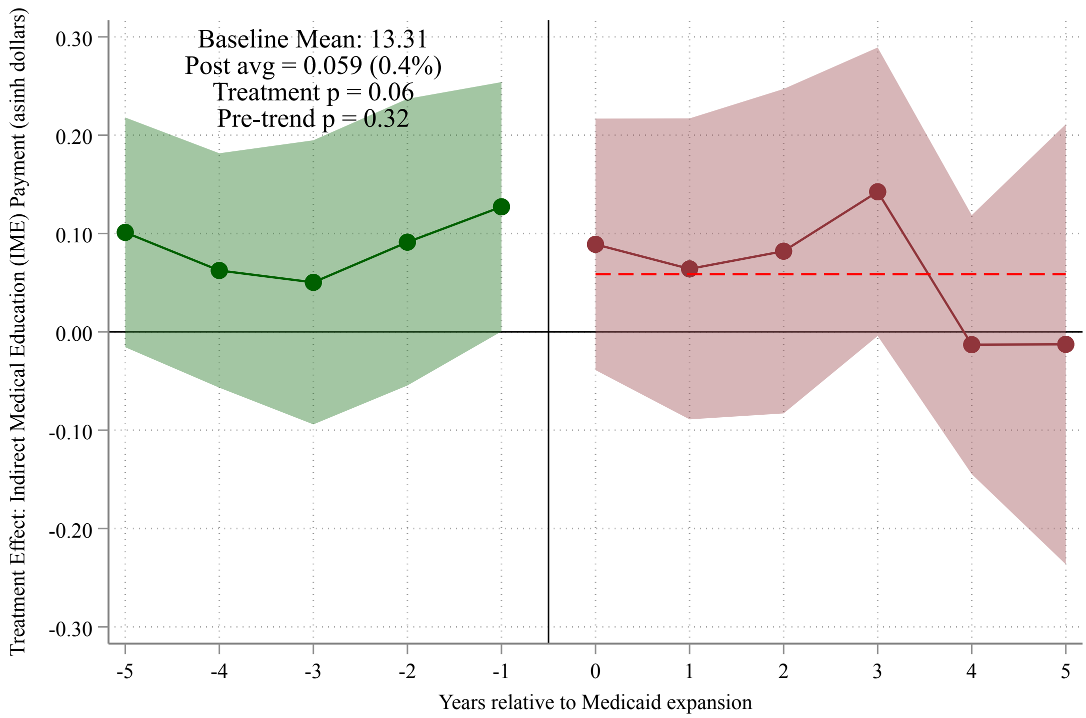

*Notes:* The figure plots event-study coefficients from the imputation estimator of Borusyak, Jaravel, and Spiess (2024), where the outcome is matched residency positions per 100,000 *contemporary* (year-varying) state population from the American Community Survey. The horizontal axis measures years relative to a state's Medicaid expansion; year 0 is the first year of expansion. Green circles (years $-5$ to $-1$) show pre-expansion coefficients. Dark red circles (years 0 to $+5$) show post-expansion treatment effects. The dashed red line indicates the average post-expansion treatment effect ($-0.016$), approximately $-7.3$ percent relative to the pre-expansion mean of 0.22 matched positions per 100,000 (joint $p = 0.10$); the pre-expansion coefficients are jointly small and consistent with parallel trends (pre-trend $p = 0.26$). As discussed in the text, the aggregate magnitude is imprecise under randomization inference; the sign, dynamics, and mechanism are the robust features. Shaded bands are 95 percent confidence intervals. Regressions include hospital and year fixed effects and weight by 2010 state population. Standard errors are clustered at the state level.

Figure 6: Effect of Medicaid Expansion on Direct GME (DGME) Payments: Event Study Estimates



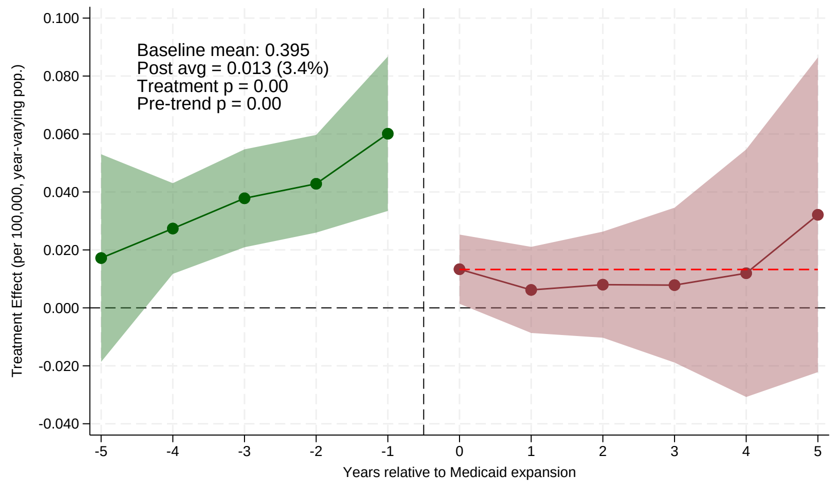

*Notes:* The figure plots event-study coefficients from the imputation estimator of Borusyak, Jaravel, and Spiess (2024). The outcome is the inverse hyperbolic sine (asinh) of Direct GME (DGME) payments per teaching hospital, from CMS Medicare hospital cost reports; DGME reimburses the direct costs of residency training (resident salaries, benefits, and faculty supervision). Because $\text{asinh}(x) \approx \ln(2x)$ for large $x$, coefficients approximate proportional changes. The sample comprises all hospitals receiving Medicare GME payments, observed at the hospital–fiscal-year level, with never-expansion states as the comparison group. The horizontal axis measures years relative to a state's Medicaid expansion; year 0 is the first year of expansion. Green circles (years $-5$ to $-1$) show pre-expansion coefficients; dark red circles (years 0 to $+5$) show post-expansion treatment effects. The dashed red line indicates the average post-expansion effect of 0.140, roughly a 14 percent increase in DGME payments (joint $p < 0.01$), with the largest effects in the first two post-expansion years. Pre-expansion coefficients are near zero from event years $-5$ to $-2$; the elevated coefficient at $-1$ (joint pre-trend $p = 0.07$) is consistent with modest anticipatory increases in payments. Shaded bands are 95 percent confidence intervals. Estimates are based on 1,916 hospitals across 51 states and the District of Columbia. Regressions include hospital and year fixed effects and are unweighted. Standard errors are clustered at the state level.

Figure 7: Effect of Medicaid Expansion on Indirect Medical Education (IME) Payments: Event Study Estimates

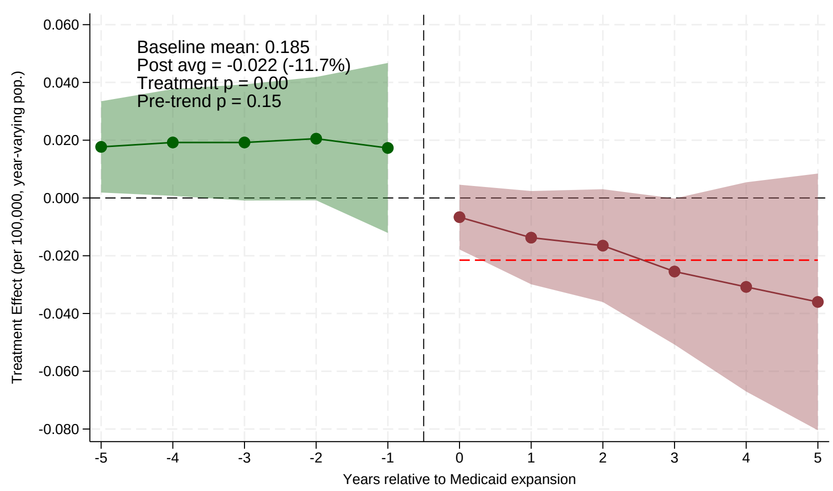



*Notes:* The figure plots event-study coefficients from the imputation estimator of Borusyak, Jaravel, and Spiess (2024) for the indirect component of GME payments, complementing the Direct GME estimate in Figure 6. The outcome is the inverse hyperbolic sine (asinh) of Indirect Medical Education (IME) payments per teaching hospital, from CMS Medicare hospital cost reports; IME is the indirect operating-cost adjustment paid to teaching hospitals, as distinct from the direct training costs captured by DGME. Because $\text{asinh}(x) \approx \ln(2x)$ for large $x$, coefficients approximate proportional changes. The average post-expansion effect is 0.059, roughly a 6 percent increase, and is only marginally significant (joint $p = 0.06$); pre-expansion coefficients are jointly insignificant (joint pre-trend $p = 0.32$). The muted response relative to Direct GME indicates that the GME-financing effect of expansion operates primarily through direct resident-training payments rather than the indirect adjustment. The sample comprises all hospitals receiving Medicare GME payments at the hospital–fiscal-year level, with never-expansion states as the comparison group. Green circles (years $-5$ to $-1$) show pre-expansion coefficients; dark red circles (years 0 to $+5$) show post-expansion treatment effects. Shaded bands are 95 percent confidence intervals. Estimates are based on 1,916 hospitals across 51 states and the District of Columbia. Regressions include hospital and year fixed effects and are unweighted. Standard errors are clustered at the state level.

Figure 8: Mechanism: Effect of Medicaid Expansion by State Medicaid GME Formula



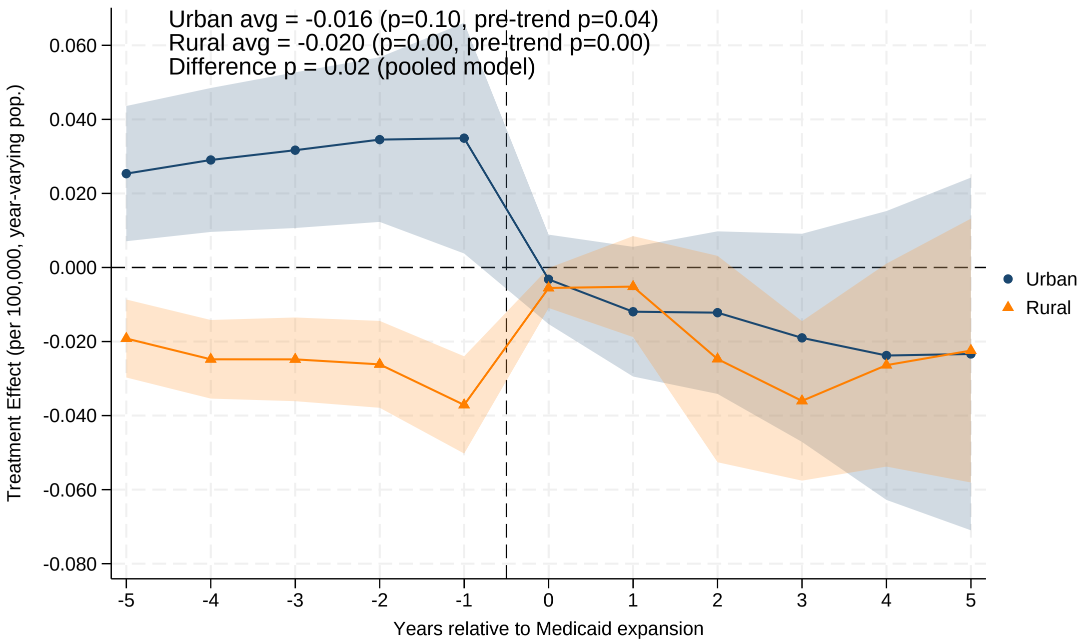

(a) Volume-Responsive GME States




(b) Non-Responsive GME States (Fixed/None)

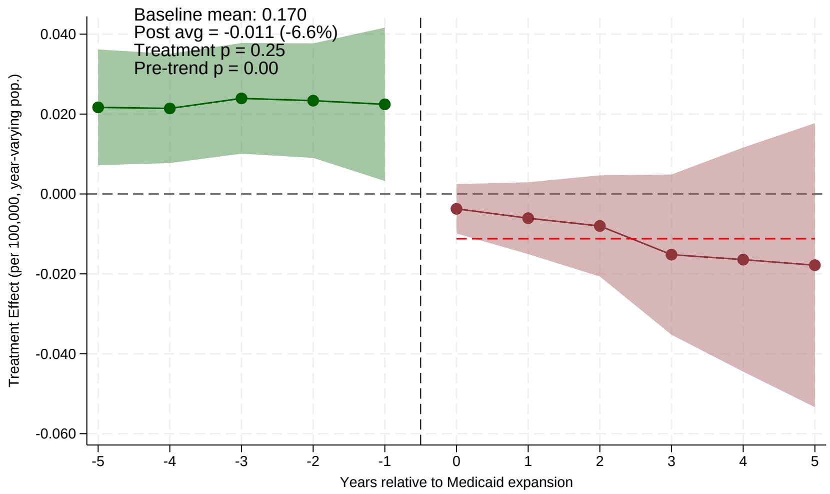
*Notes:* The figure plots event-study coefficients from the imputation estimator of Borusyak, Jaravel, and Spiess (2024), estimated separately for expansion states with volume-responsive (top) versus non-responsive (bottom) Medicaid GME formulas, each using non-expansion states as the comparison group. The outcome is matched residency positions per 100,000 contemporary (year-varying) state population. In non-responsive states the average post-expansion effect is $-0.022$ (an 11.7 percent decline; joint $p < 0.01$) following flat pre-expansion coefficients (pre-trend $p = 0.15$); in volume-responsive states it is $+0.013$ ($+3.4$ percent), with no decline. A pooled interacted model puts the volume-minus-non-responsive difference in average post effects at 0.035 per 100,000 (clustered $p < 0.001$; randomization-inference $p = 0.19$). The volume-responsive panel exhibits a pronounced upward pre-trend (pre-trend $p < 0.001$), so its estimate should be read with caution; the non-responsive contraction is the more robust leg of the comparison. Volume-responsive states reward higher Medicaid volume with larger GME payments, so expansion mechanically raises GME revenue; non-responsive states (fixed per-resident amounts, lump-sum grants, or no Medicaid GME) gain none. States are classified from the Medicaid GME payment methodology reported in Henderson (2013). Green circles (years $-5$ to $-1$) show pre-expansion coefficients; dark red circles (years 0 to $+5$) show post-expansion treatment effects; the dashed red line marks the average post-expansion effect. Shaded bands are 95 percent confidence intervals. Regressions include hospital and year fixed effects, are weighted by state population. Standard errors are clustered at the state level.

Figure 9: Effect of Medicaid Expansion on Matched Residency Positions by Hospital Location: Event Study Estimates

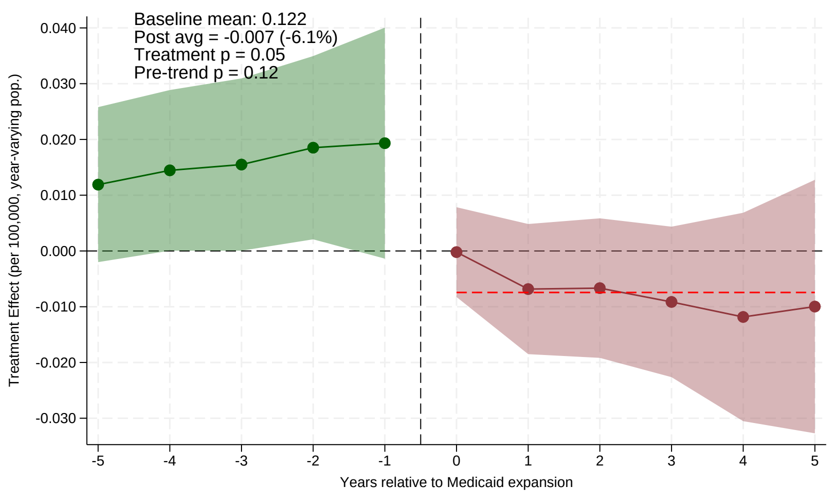



*Notes:* The figure plots event-study coefficients from the imputation estimator of Borusyak, Jaravel, and Spiess (2024), estimated in two split-sample regressions for urban (blue circles) and rural (orange triangles) hospitals; each subsample keeps the never-expansion hospitals of its own type as the comparison group, so each series carries its own pre-expansion path and pre-trends test. Urban hospitals are those in counties with 2010 RUCA codes 1–3; rural hospitals are those with RUCA codes above 3. The outcome is matched positions per 100,000 contemporary (year-varying) population. The horizontal axis measures years relative to a state's Medicaid expansion; year 0 is the first year of expansion. The average post-expansion treatment effect is $-0.016$ for urban hospitals (joint $p = 0.10$; pre-trend $p = 0.04$, with uniformly positive pre-expansion coefficients) and $-0.020$ for rural hospitals (joint $p < 0.001$). The rural subsample fails its own pre-trends test ($p < 0.001$), with negative pre-expansion coefficients comparable in size to the post-expansion effects, so the rural series should be read descriptively. A pooled interacted model puts the urban–rural difference in average post effects at $-0.014$ ($p = 0.02$), but that model shares pre-trends across groups. Shaded bands are 95 percent confidence intervals. Regressions include hospital and year fixed effects and weight by 2010 state population. Standard errors are clustered at the state level.

Figure 10: Effect of Medicaid Expansion on Matched Residency Positions by Specialty Type: Event Study Estimates



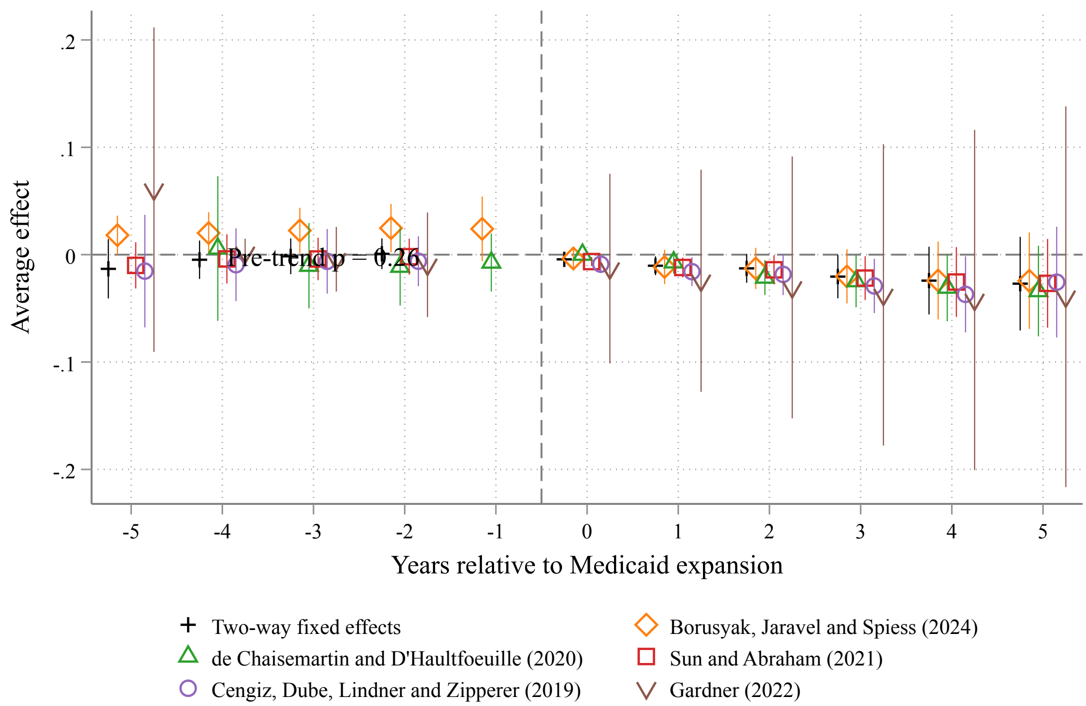

(a) Non-Primary Care Positions




(b) Primary Care Positions

*Notes:* The figure plots event-study coefficients from the imputation estimator of Borusyak, Jaravel, and Spiess (2024) estimated separately for non-primary care (top panel) and primary care specialties (bottom panel). Primary care includes family medicine, internal medicine, and pediatrics. The outcome is matched positions per 100,000 contemporary (year-varying) population. The horizontal axis measures years relative to a state's Medicaid expansion; year 0 is the first year of expansion. For non-primary care programs, the average post-expansion treatment effect is $-0.002$ ($-4.7$ percent; $p = 0.22$; pre-trend $p = 0.01$). For primary care programs, it is $-0.003$ ($-4.6$ percent; $p = 0.06$; pre-trend $p = 0.06$). Both panels exhibit differential pre-trends, so the specialty estimates should be read as suggestive. Shaded bands are 95 percent confidence intervals. Regressions include hospital and year fixed effects and weight by 2010 state population. Standard errors are clustered at the state level.

Figure 11: Robustness of Event Study Estimates Across Difference-in-Differences Estimators: Matched Residency Positions per 100,000 Population




*Notes:* The figure compares event-study estimates of the effect of Medicaid expansion on matched residency positions per 100,000 contemporary (year-varying) population across six difference-in-differences estimators that are robust to treatment-effect heterogeneity under staggered adoption: two-way fixed effects; Borusyak, Jaravel, and Spiess (2024); de Chaisemartin and D'Haultfoeuille (2020); Sun and Abraham (2021); Cengiz, Dube, Lindner, and Zipperer (2019); and Gardner (2022). The heterogeneity-robust estimator of Callaway and Sant'Anna (2021) is excluded because it is not well identified in the sparsely populated never-treated cells of this sample. The pre-trend p-value reported in the panel is from the Borusyak, Jaravel, and Spiess (2024) estimator. The horizontal axis measures years relative to a state's Medicaid expansion; the dashed vertical line marks the expansion year. Regressions include hospital and year fixed effects and weight by 2010 state population. Standard errors are clustered at the state level.

ONLINE APPENDIX

# Medicaid Expansion and the Reallocation of Medical Residency Training

Hussain Hadah Ricardo B. Ang III

# A Tables

Table A.1: Multiple-Testing Adjustment for the Primary Family of Per-Capita Effects

| | Clustered | | Randomization inference | |
|---|---|---|---|---|
| Hypothesis (avg. post effect, matched per 100,000) | p-value | q-value | p-value | q-value |
| Full sample (headline) | 0.103 | 0.123 | 0.579 | 0.762 |
| Urban hospitals | 0.095 | 0.123 | 0.635 | 0.762 |
| Rural hospitals | <0.001 | 0.002 | 0.289 | 0.762 |
| Primary care | 0.063 | 0.123 | 0.762 | 0.762 |
| Non-primary care | 0.219 | 0.219 | 0.475 | 0.762 |
| Offered positions (quota) | 0.033 | 0.098 | 0.515 | 0.762 |

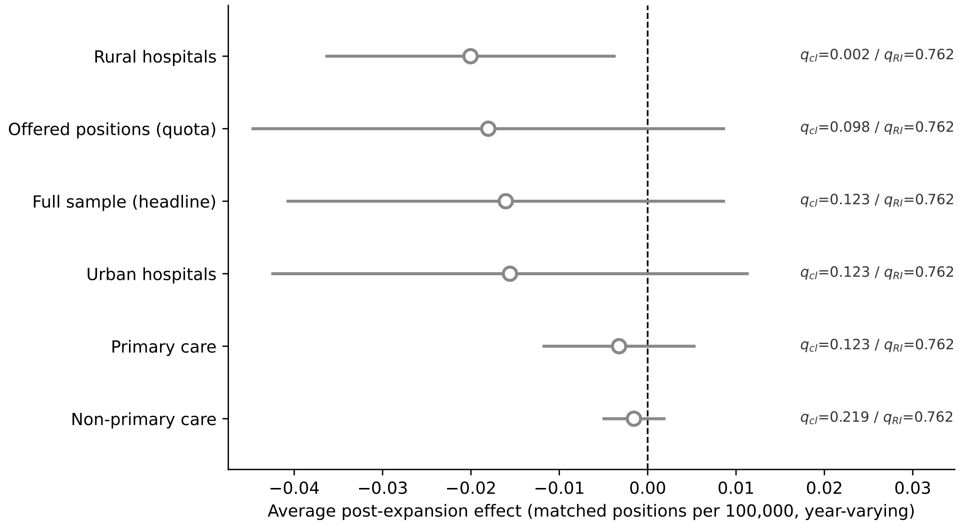
*Notes:* The table reports Benjamini–Hochberg false-discovery-rate q-values for the primary family of average post-expansion effects on matched positions per 100,000 population, computed under two inference standards. “Clustered” p-values are the joint tests of the post-expansion event-time coefficients with standard errors clustered at the state level (the offered-positions row uses the quota outcome). “Randomization inference” p-values permute the vector of state expansion-timing cohorts across the states in each specification’s sample under the sharp null of no effect (1,000 permutation draws for the full-sample headline, 500 for the other members) and compare the observed average post effect with the permutation distribution; this is the conservative standard the text leads with, applied uniformly to every family member. Within each standard, q-values apply the Benjamini–Hochberg correction across the family of six tests; a q-value below 0.05 indicates a finding that survives a 5 percent false-discovery-rate correction.

Table A.2: Baseline Balance Across Medicaid GME Formula Groups (2010)

| Baseline characteristic (2010) | Volume-responsive (1) | Fixed/none (2) | Never-expansion (3) | Difference (1)−(2) | p-value |
|---|---|---|---|---|---|
| Matched positions per program | 26.4 | 29.4 | 30.0 | -3.0 | 0.576 |
| Offered positions (quota) per program | 28.0 | 30.9 | 31.6 | -2.8 | 0.604 |
| Share of programs filling every offered position | 0.81 | 0.79 | 0.87 | 0.02 | 0.767 |
| Matched positions per 100,000 population | 10.25 | 8.26 | 5.46 | 1.99 | 0.594 |
| Program-level mean, population-weighted | 0.36 | 0.17 | 0.24 | 0.19 | – |
| Offered positions (quota) per 100,000 | 10.80 | 8.64 | 5.76 | 2.16 | 0.571 |
| Number of programs | 13.7 | 23.8 | 12.5 | -10.1 | 0.198 |
| State population (millions) | 4.2 | 7.4 | 6.3 | -3.2 | 0.177 |
| Rural program share | 0.10 | 0.08 | 0.10 | 0.02 | 0.625 |
| Primary-care share of matched positions | 0.52 | 0.55 | 0.57 | -0.03 | 0.478 |
| Mean expansion year | 2014.5 | 2014.5 | – | 0.0 | 1.000 |
| Number of states | 16 | 18 | 17 | | |

*Notes:* The table reports baseline (2010, pre-treatment) means by Medicaid GME formula group, for the same characteristics as the summary-statistics table (Table 1). Expansion states are classified as volume-responsive or fixed/none from the payment methodology reported in the 2012 survey of Henderson (2013); never-expansion states are shown for reference. All rows except the population-weighted row are unweighted means across states (the state is the unit of observation); the p-value is from a Welch two-sample t-test of the volume-responsive versus fixed/none difference at the state level, so power is limited by the number of states. The population-weighted row reports the program-level mean of matched positions per 100,000 (2010 population), weighted by state population—the object the estimation weights by; its cross-group gap reflects the concentration of training in a few large volume-responsive states rather than state-level imbalance. The fill-share row is measured at the NRMP specialty-program level (the share of a state's specialty programs filling every offered position); the primary-care share is the share of matched positions in family medicine, internal medicine, and pediatrics; rural programs are those in counties with 2010 RUCA codes above 3. Data are from NRMP Main Residency Match reports and the U.S. Census Bureau.

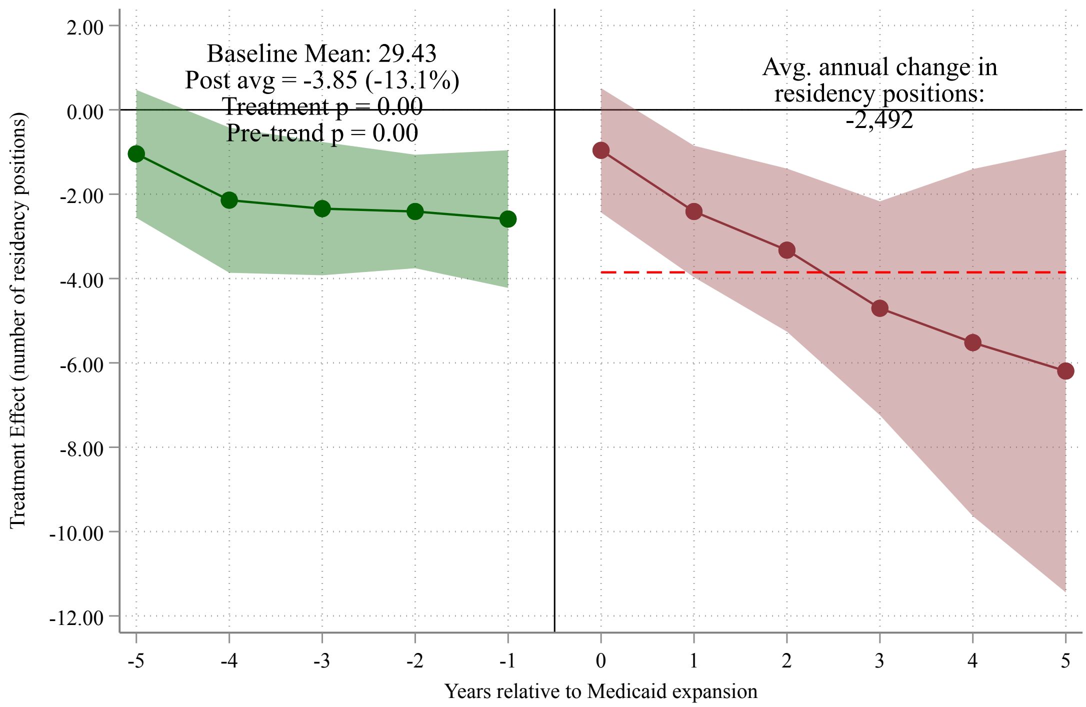
Figure A.1: Average Post-Expansion Effects and False-Discovery-Rate Significance Across the Primary Family




*Notes:* The figure plots the average post-expansion effect on matched positions per 100,000 contemporary (year-varying) population for each member of the primary family of hypotheses, with 95 percent confidence intervals (point estimate $\pm 1.96$ clustered standard errors). The label beside each marker reports the Benjamini–Hochberg false-discovery-rate q-value under both inference standards: $q_{cl}$ (clustered) and $q_{RI}$ (randomization inference). Filled markers would denote effects surviving a 5 percent false-discovery-rate correction under the conservative randomization standard; no member survives it ($q_{RI} \geqslant 0.76$). Under clustered inference the rural effect is the lone survivor ($q = 0.002$), a result we discount because the rural subsample fails its own pre-trends test (see Section 6.4). Standard errors are clustered at the state level.

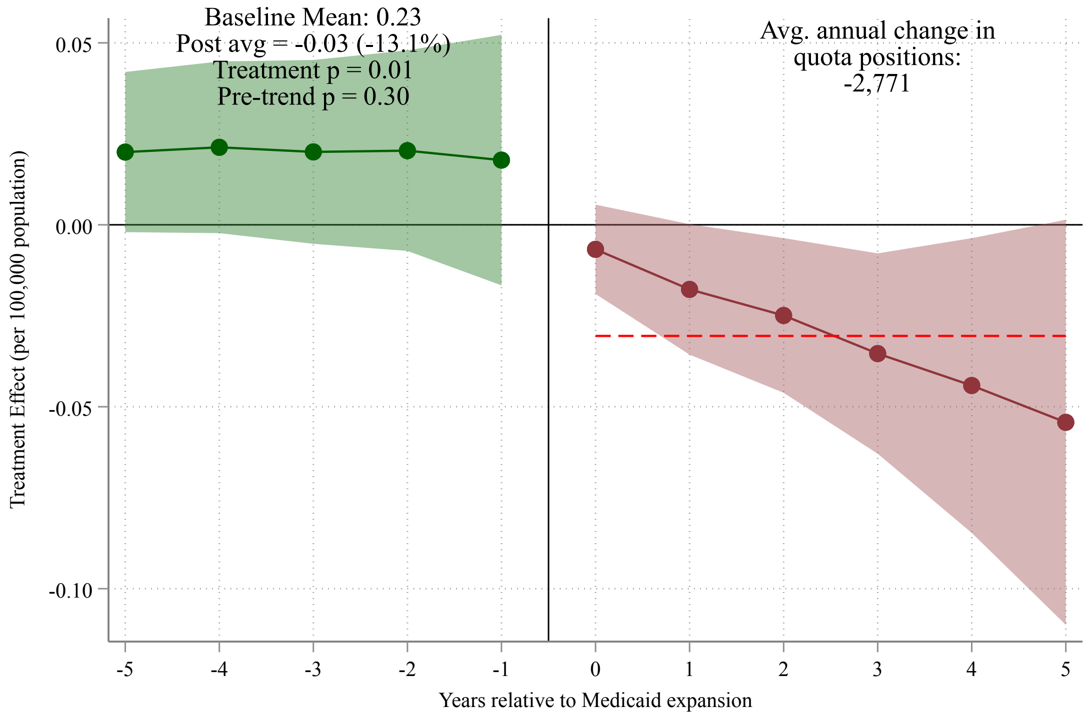
# B Figures

Figure A.2: Effect of Medicaid Expansion on Matched Residency Positions in Levels: Event Study Estimates



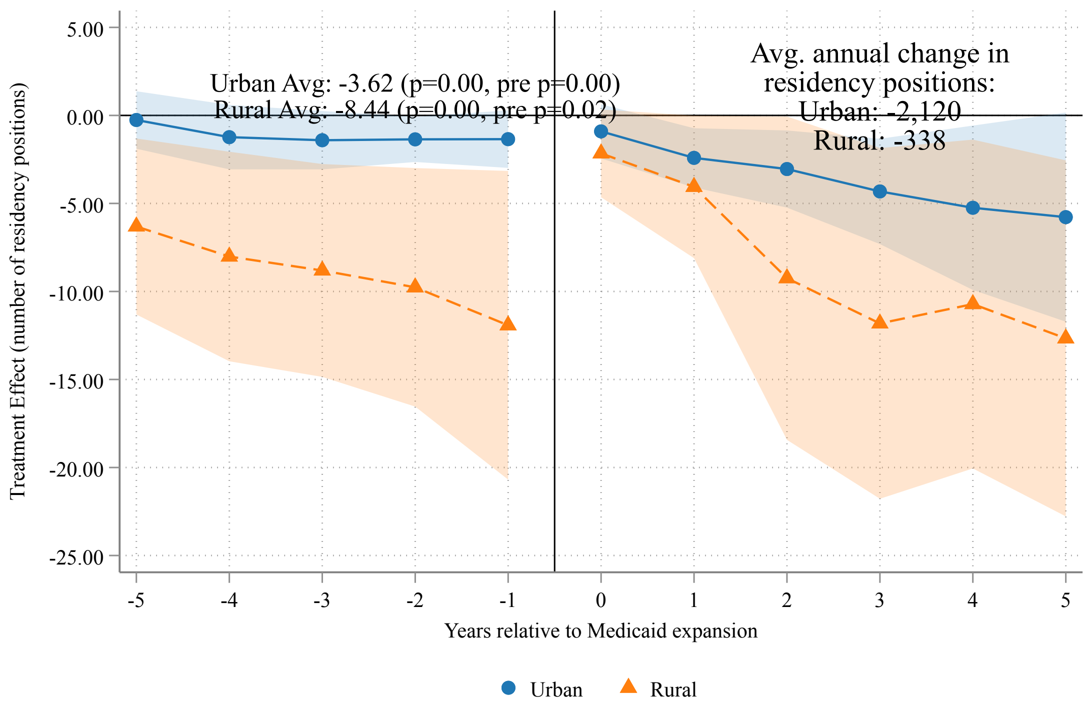

*Notes:* The figure plots event-study coefficients from the imputation estimator of Borusyak, Jaravel, and Spiess (2024), where the outcome is the absolute number of matched residency positions per hospital (not normalized by population). The horizontal axis measures years relative to a state's Medicaid expansion; year 0 is the first year of expansion. Green circles (years $-5$ to $-1$) show pre-expansion coefficients; dark red circles (years 0 to +5) show post-expansion treatment effects. The dashed red line indicates the average post-expansion treatment effect of $-3.85$ positions per hospital, corresponding to a 13.1 percent decline relative to the pre-expansion mean of 29.43 matched positions ($p < 0.01$). **The pre-expansion coefficients in this raw-count specification are large and negative** (on the order of $-2$ positions per hospital, comparable in magnitude to the treatment effect), indicating a violation of parallel trends. This levels estimate therefore corroborates the *direction* of the effect but should not be read as a clean causal magnitude; the paper's primary specification instead normalizes by contemporary population, which restores flat pre-trends (see Section 7.3). Shaded bands are 95 percent confidence intervals. Regressions include hospital and year fixed effects. Standard errors are clustered at the state level.

Figure A.3: Effect of Medicaid Expansion on Offered Residency Positions (Quota): Event Study Estimates



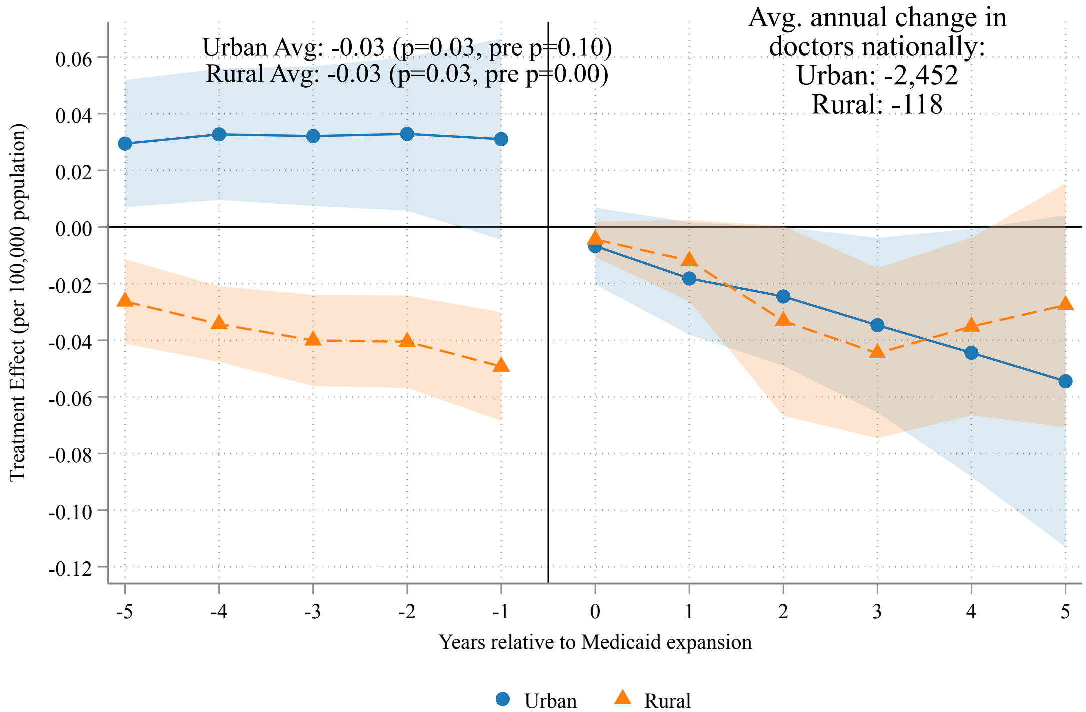

*Notes:* The figure plots event-study coefficients from the imputation estimator of Borusyak, Jaravel, and Spiess (2024), where the outcome is quota positions per 100,000 state population (fixed 2010 population; the year-varying quota estimate of $-0.018$ is reported in the text). Quota positions are the number of positions offered by residency programs in the NRMP Main Residency Match, prior to the matching algorithm; matched positions (the primary outcome in Figure 5) are the subset of offered positions that are ultimately filled. The average post-expansion treatment effect is $-0.03$ ($-13.1$ percent relative to the pre-expansion mean of 0.23 offered positions per 100,000; $p = 0.01$), corresponding to an average annual reduction of approximately 2,771 offered positions nationally. Shaded bands are 95 percent confidence intervals. Regressions include hospital and year fixed effects. Standard errors are clustered at the state level.

Figure A.4: Effect of Medicaid Expansion on Matched Residency Positions by Hospital Location, in Levels: Event Study Estimates



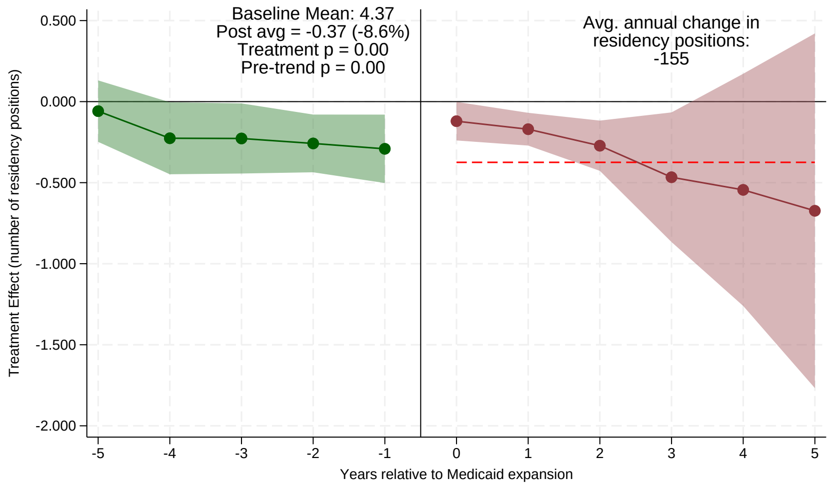

*Notes:* The figure plots event-study coefficients from the imputation estimator of Borusyak, Jaravel, and Spiess (2024), estimated in two split-sample regressions for urban (blue circles) and rural (orange triangles) hospitals; each subsample keeps the never-expansion hospitals of its own type as the comparison group, so each series carries its own pre-expansion path and pre-trends test. The outcome is the absolute number of matched residency positions per hospital (not normalized by population). The average post-expansion treatment effect is $-3.62$ positions per hospital for urban hospitals (joint $p < 0.001$; pre-trend $p < 0.001$) and $-8.44$ positions per hospital for rural hospitals (joint $p < 0.001$; pre-trend $p = 0.02$). As with the aggregate levels outcome (Figure A.2), both subsamples fail the pre-trends test in levels, so these magnitudes corroborate the direction of the effect only. Translating to aggregate counts, expansion states experienced average annual reductions of approximately 2,120 matched positions at urban hospitals and 338 at rural hospitals. Shaded bands are 95 percent confidence intervals. Regressions include hospital and year fixed effects and weight by 2010 state population. Standard errors are clustered at the state level.

Figure A.5: Effect of Medicaid Expansion on Offered Residency Positions (Quota) by Hospital Location: Event Study Estimates

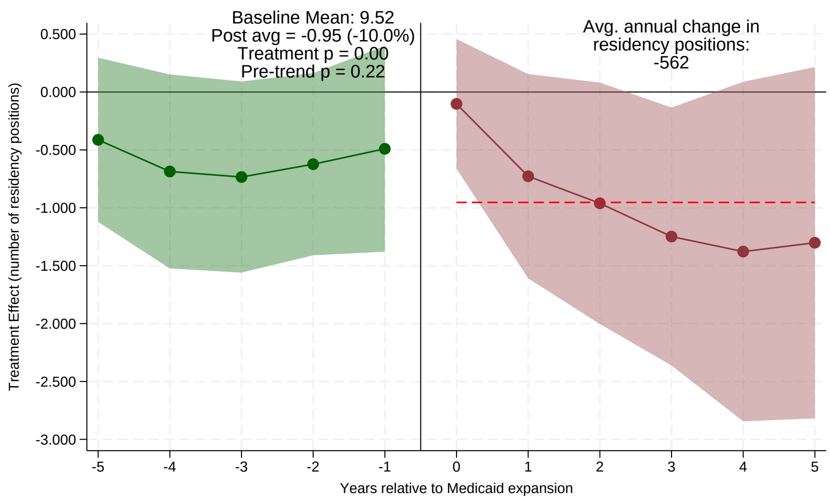



*Notes:* The figure plots event-study coefficients from the imputation estimator of Borusyak, Jaravel, and Spiess (2024), estimated in two split-sample regressions for urban (blue circles) and rural (orange triangles) hospitals; each subsample keeps the never-expansion hospitals of its own type as the comparison group, so each series carries its own pre-expansion path and pre-trends test. The outcome is quota positions per 100,000 state population (fixed 2010 population), where quota measures positions offered by programs prior to the matching algorithm. The average post-expansion treatment effect is $-0.031$ for urban hospitals (joint $p = 0.03$; pre-trend $p = 0.10$) and $-0.026$ for rural hospitals (joint $p = 0.03$), similar in magnitude across the two groups. The rural subsample fails its own pre-trends test ($p < 0.001$), so the rural series should be read descriptively. These estimates correspond to average annual reductions of approximately 2,452 offered positions at urban hospitals and 118 at rural hospitals. Shaded bands are 95 percent confidence intervals. Regressions include hospital and year fixed effects and weight by 2010 state population. Standard errors are clustered at the state level.

Figure A.6: Effect of Medicaid Expansion on Matched Residency Positions by Specialty Type, in Levels: Event Study Estimates



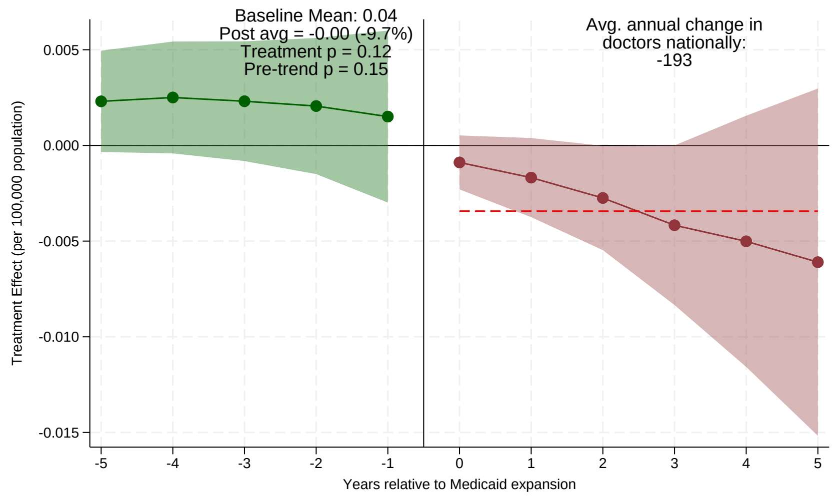

(a) Non-Primary Care Positions



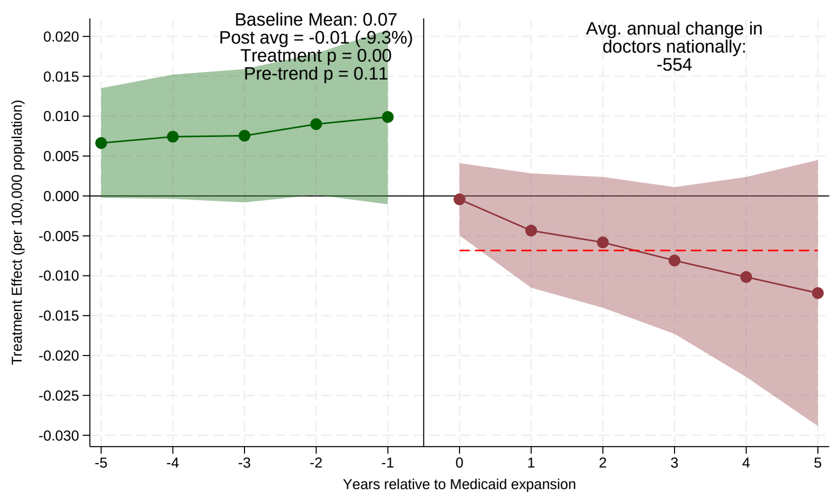

(b) Primary Care Positions

*Notes:* The figure plots event-study coefficients from the imputation estimator of Borusyak, Jaravel, and Spiess (2024), estimated separately for non-primary care (top panel) and primary care specialties (bottom panel). The outcome is the absolute number of matched residency positions per hospital (not normalized by population). Primary care includes family medicine, internal medicine, and pediatrics. The average post-expansion treatment effect is $-0.37$ positions per hospital for non-primary care ($-8.6$ percent relative to the pre-expansion mean of 4.37; $p < 0.01$), corresponding to an average annual reduction of approximately 155 matched positions nationally. For primary care, the average treatment effect is $-0.95$ positions per hospital ($-10.0$ percent relative to the pre-expansion mean of 9.52; $p < 0.01$), corresponding to an average annual reduction of approximately 562 positions. Shaded bands are 95 percent confidence intervals. Regressions include hospital and year fixed effects. Standard errors are clustered at the state level.

Figure A.7: Effect of Medicaid Expansion on Offered Residency Positions (Quota) by Specialty Type: Event Study Estimates



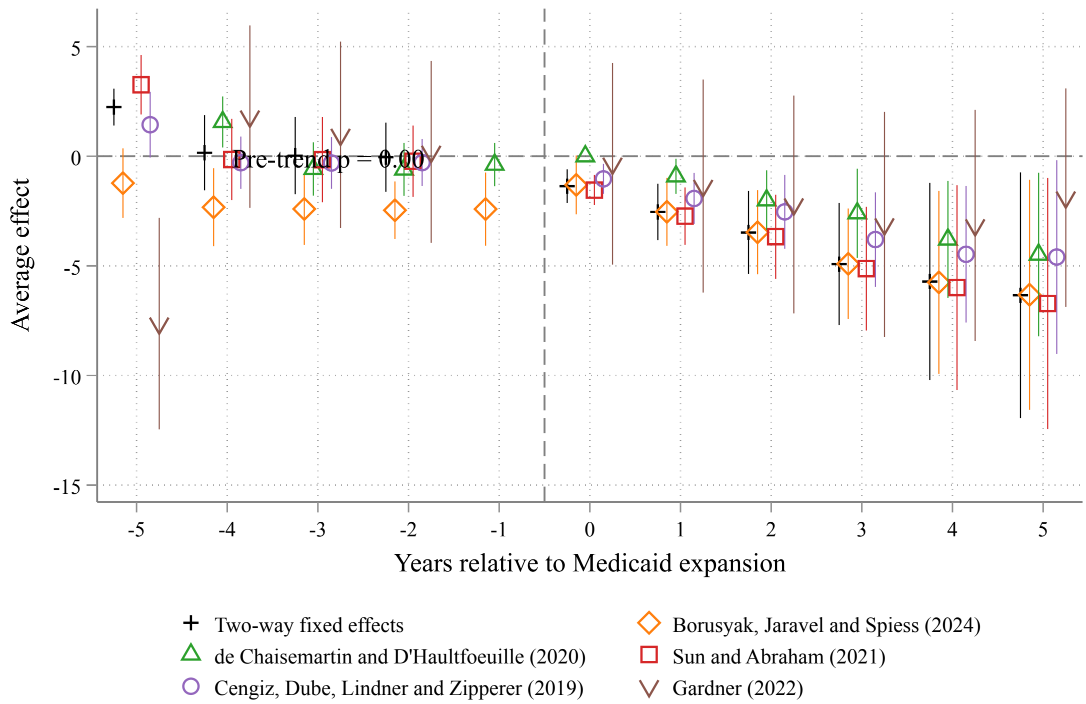

(a) Non-Primary Care Positions




(b) Primary Care Positions

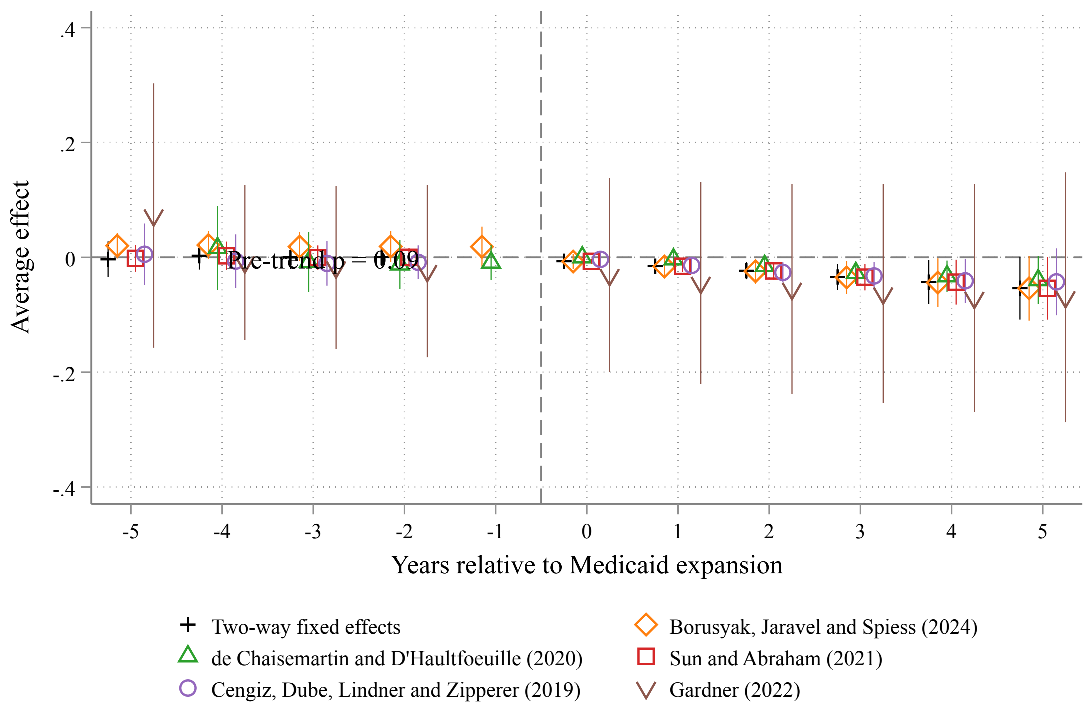
*Notes:* The figure plots event-study coefficients from the imputation estimator of Borusyak, Jaravel, and Spiess (2024), estimated separately for non-primary care (top panel) and primary care specialties (bottom panel). The outcome is quota positions per 100,000 state population, where quota measures positions offered by programs prior to the matching algorithm. For non-primary care, the average post-expansion treatment effect is $-0.00$ ($-9.7$ percent relative to the pre-expansion mean of 0.04; $p = 0.12$), corresponding to an average annual reduction of approximately 193 offered positions nationally. For primary care, the average treatment effect is $-0.01$ ($-9.3$ percent relative to the pre-expansion mean of 0.07; $p < 0.01$), corresponding to an average annual reduction of approximately 554 offered positions. Shaded bands are 95 percent confidence intervals. Regressions include hospital and year fixed effects. Standard errors are clustered at the state level.

Figure A.8: Robustness of Event Study Estimates Across Difference-in-Differences Estimators, Total Matched Residency Positions in Levels



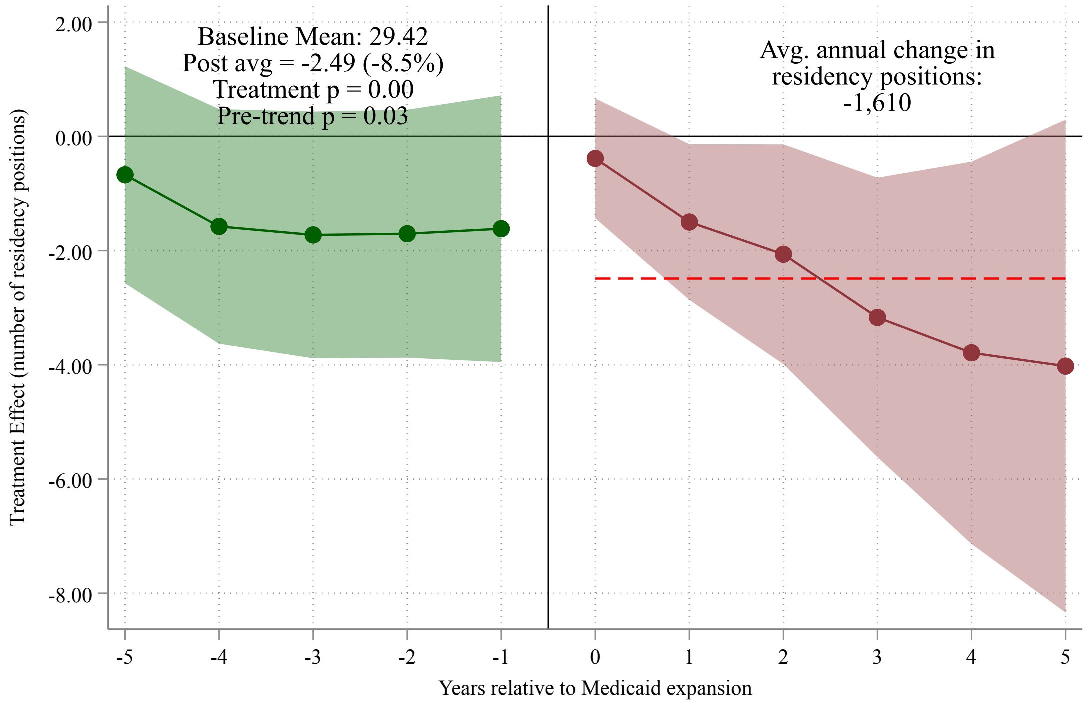

*Notes:* The figure compares event-study estimates of the effect of Medicaid expansion on the absolute number of matched residency positions per hospital across six difference-in-differences estimators that are robust to treatment-effect heterogeneity under staggered adoption: two-way fixed effects; Borusyak, Jaravel, and Spiess (2024); de Chaisemartin and D'Haultfoeuille (2020); Sun and Abraham (2021); Cengiz, Dube, Lindner, and Zipperer (2019); and Gardner (2022). The heterogeneity-robust estimator of Callaway and Sant'Anna (2021) is excluded because it is not well identified in the sparsely populated never-treated cells of this sample. The pre-trend p-value reported in the panel is from the Borusyak, Jaravel, and Spiess (2024) estimator. The dashed vertical line marks the expansion year. Regressions include hospital and year fixed effects. Standard errors are clustered at the state level.

Figure A.9: Robustness of Event Study Estimates Across Difference-in-Differences Estimators, Residency Quota Positions per 100,000 Population



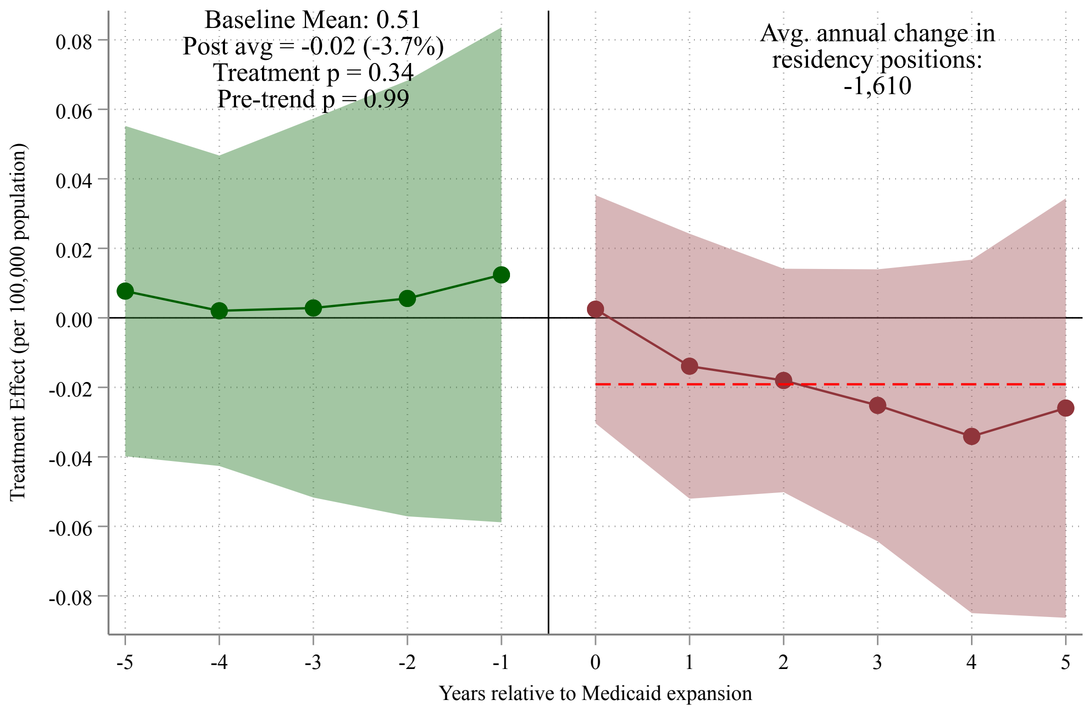

*Notes:* The figure compares event-study estimates of the effect of Medicaid expansion on quota positions per 100,000 state population across six difference-in-differences estimators. Quota positions measure positions offered by residency programs prior to the matching algorithm. The estimators are: two-way fixed effects; Borusyak, Jaravel, and Spiess (2024); de Chaisemartin and D'Haultfoeuille (2020); Sun and Abraham (2021); Cengiz, Dube, Lindner, and Zipperer (2019); and Gardner (2022); the Callaway and Sant'Anna (2021) estimator is excluded as in the previous figure. The dashed vertical line marks the expansion year. Regressions include hospital and year fixed effects. Standard errors are clustered at the state level.

Figure A.10: Effect of Medicaid Expansion on Matched Residency Positions (Unweighted): Event Study Estimates



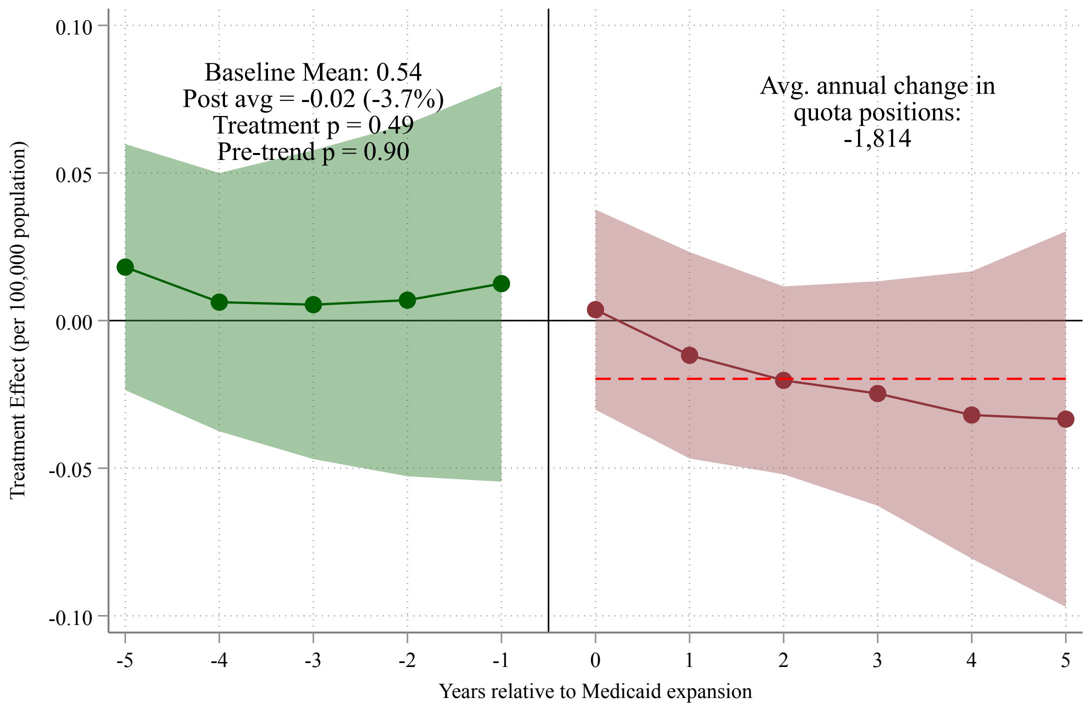

*Notes:* The figure plots event-study coefficients from the imputation estimator of Borusyak, Jaravel, and Spiess (2024), where the outcome is the number of matched residency positions. This specification is estimated without population weights and includes 2010 state population as a control. The horizontal axis measures years relative to a state's Medicaid expansion; year 0 is the first year of expansion. Green circles (years $-5$ to $-1$) show pre-expansion coefficients. Dark red circles (years 0 to $+5$) show post-expansion treatment effects. The dashed red line indicates the average post-expansion treatment effect ($-2.49$), which corresponds to approximately $-8.5$ percent relative to the pre-expansion mean of 29.42 matched positions (p-value $=$ 0.00). Translating to aggregate counts, expansion states experienced an average annual reduction of approximately 1,610 matched residency positions relative to the counterfactual. Shaded bands are 95 percent confidence intervals. Regressions include hospital and year fixed effects. Standard errors are clustered at the state level.

Figure A.11: Effect of Medicaid Expansion on Matched Residency Positions per 100,000 Population (Unweighted): Event Study Estimates



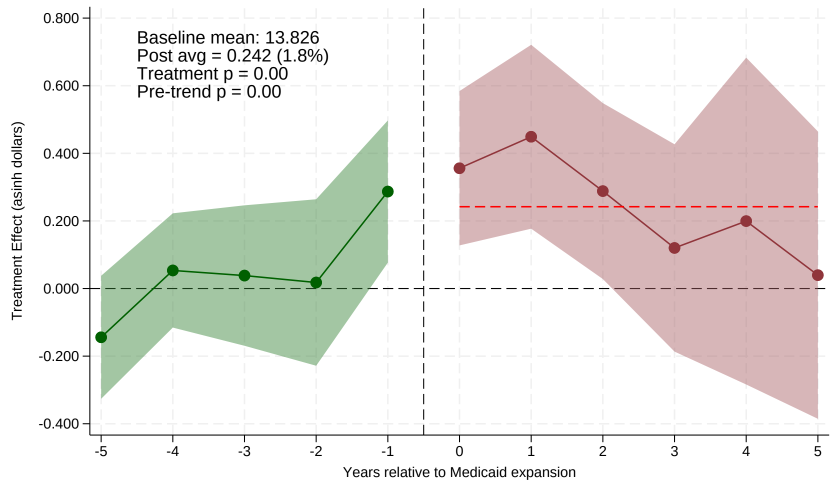

*Notes:* The figure plots event-study coefficients from the imputation estimator of Borusyak, Jaravel, and Spiess (2024), where the outcome is matched residency positions per 100,000 state population. This specification is estimated without population weights and includes 2010 state population as a control. The horizontal axis measures years relative to a state's Medicaid expansion; year 0 is the first year of expansion. Green circles (years −5 to −1) show pre-expansion coefficients. Dark red circles (years 0 to +5) show post-expansion treatment effects. The dashed red line indicates the average post-expansion treatment effect (−0.02), which corresponds to approximately −3.7 percent relative to the pre-expansion mean of 0.51 matched positions per 100,000 (p-value = 0.34). Translating to aggregate counts, expansion states experienced an average annual reduction of approximately 1,610 matched residency positions relative to the counterfactual. Shaded bands are 95 percent confidence intervals. Regressions include hospital and year fixed effects. Standard errors are clustered at the state level.

Figure A.12: Effect of Medicaid Expansion on Residency Quota Positions per 100,000 Population (Unweighted): Event Study Estimates

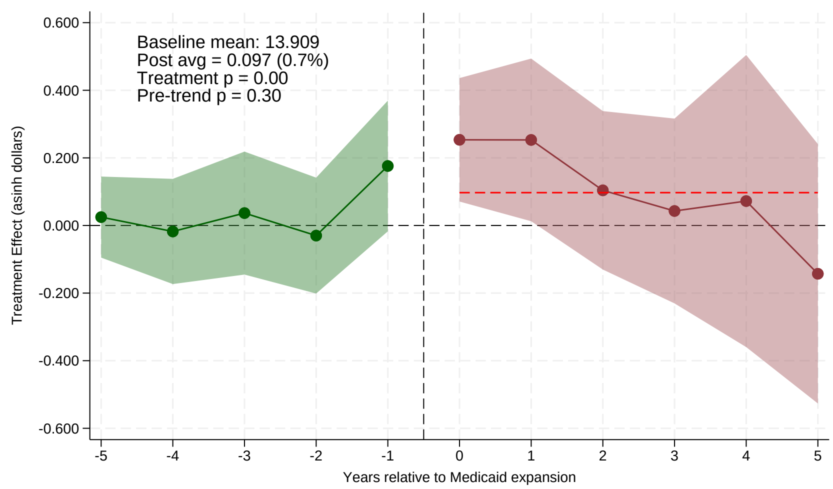



*Notes:* The figure plots event-study coefficients from the imputation estimator of Borusyak, Jaravel, and Spiess (2024), where the outcome is residency quota (offered) positions per 100,000 state population. This specification is estimated without population weights and includes 2010 state population as a control. The horizontal axis measures years relative to a state's Medicaid expansion; year 0 is the first year of expansion. Green circles (years $-5$ to $-1$) show pre-expansion coefficients. Dark red circles (years 0 to $+5$) show post-expansion treatment effects. The dashed red line indicates the average post-expansion treatment effect ($-0.02$), which corresponds to approximately $-3.7$ percent relative to the pre-expansion mean of 0.54 quota positions per 100,000 (p-value = 0.49). Translating to aggregate counts, expansion states experienced an average annual reduction of approximately 1,814 quota residency positions relative to the counterfactual. Shaded bands are 95 percent confidence intervals. Regressions include hospital and year fixed effects. Standard errors are clustered at the state level.

Figure A.13: Group-Specific Payment Evidence: Effect of Medicaid Expansion on Direct GME (DGME) Payments by State Medicaid GME Formula



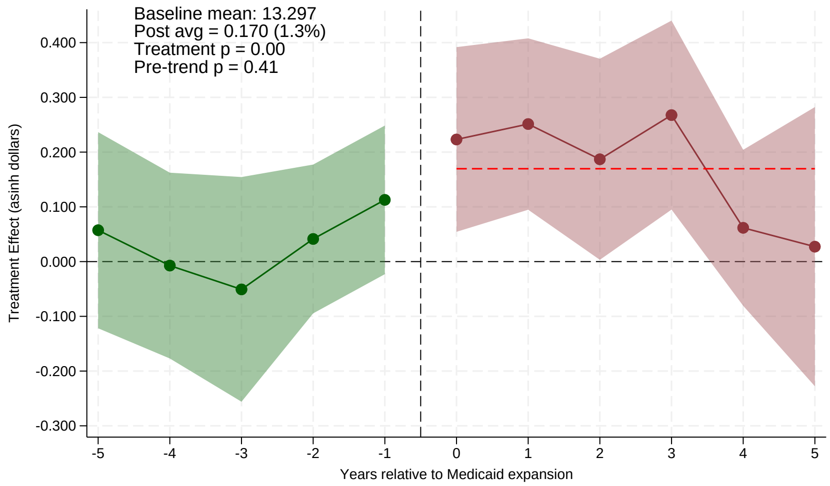

(a) Volume-Responsive GME States



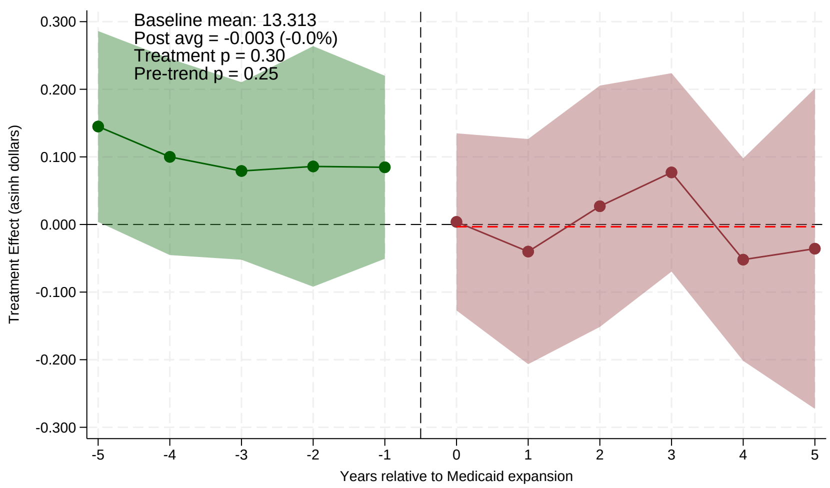

(b) Non-Responsive GME States (Fixed/None)

*Notes:* The figure re-estimates the Direct GME (DGME) payment event study from Figure 6 separately for volume-responsive (top) and non-responsive (bottom) expansion states, each using never-expansion states as the comparison group. The outcome is Medicare cost-report payments, which we read as a proxy for training-related revenue rather than a direct measure of state Medicaid GME payments (see Section 6.3). The outcome is the inverse hyperbolic sine (asinh) of DGME payments per teaching hospital, so coefficients approximate proportional changes. In volume-responsive states the average post-expansion effect is 0.24 (s.e. 0.14; roughly a 24 percent increase; joint $p < 0.01$), but this panel fails its pre-trend test (pre-trend $p = 0.0002$), so its level should be read with caution; in non-responsive states the effect is 0.10 (s.e. 0.12; roughly 10 percent; joint $p < 0.01$; pre-trend $p = 0.30$), less than half as large, though the cross-group difference is imprecise (0.14, s.e. 0.10, $p = 0.15$ from a pooled interacted model); see Figures A.15 and A.16 for the FTE/rate decomposition, where the cross-group contrast is sharper. Green circles (years $-5$ to $-1$) show pre-expansion coefficients; dark red circles (years 0 to $+5$) show post-expansion effects; the dashed red line marks the average post-expansion effect. Shaded bands are 95 percent confidence intervals. Estimates come from CMS Medicare hospital cost reports at the hospital–fiscal-year level. Regressions include hospital and year fixed effects and are unweighted. Standard errors are clustered at the state level.

Figure A.14: Group-Specific Payment Evidence: Effect of Medicaid Expansion on Indirect Medical Education (IME) Payments by State Medicaid GME Formula



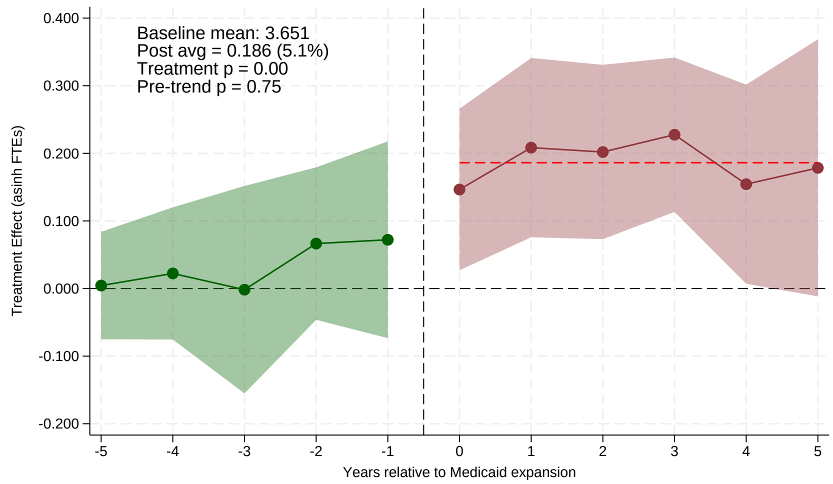

(a) Volume-Responsive GME States



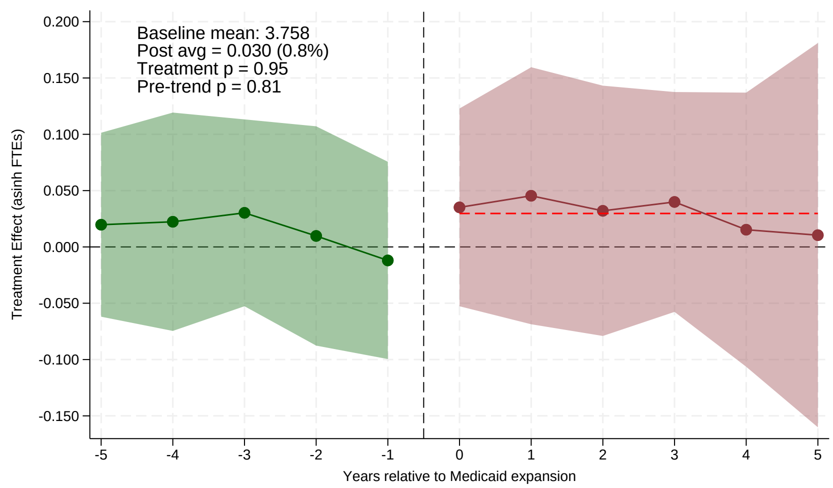

(b) Non-Responsive GME States (Fixed/None)

*Notes:* The figure re-estimates the Indirect Medical Education (IME) payment event study from Figure 7 separately for volume-responsive (top) and non-responsive (bottom) expansion states, each using never-expansion states as the comparison group. The outcome is the inverse hyperbolic sine (asinh) of IME payments per teaching hospital, from Medicare cost reports (a proxy for training-related revenue; see Section 6.3). In volume-responsive states the average post-expansion effect is 0.17 (s.e. 0.08; roughly a 17 percent increase; joint $p < 0.01$; pre-trend $p = 0.41$); in non-responsive states it is economically zero and statistically insignificant ($-0.003$, s.e. 0.07; joint $p = 0.30$; pre-trend $p = 0.25$); the cross-group difference is significant (0.15, s.e. 0.05, $p = 0.001$ from a pooled interacted model). The differential training-specific revenue gain from expansion is thus concentrated in the volume-responsive states, where residency positions do not contract. Green circles (years $-5$ to $-1$) show pre-expansion coefficients; dark red circles (years 0 to $+5$) show post-expansion effects. Shaded bands are 95 percent confidence intervals. Regressions include hospital and year fixed effects and are unweighted. Standard errors are clustered at the state level.

Figure A.15: Decomposition of the DGME Payment Response: Resident FTEs by State Medicaid GME Formula



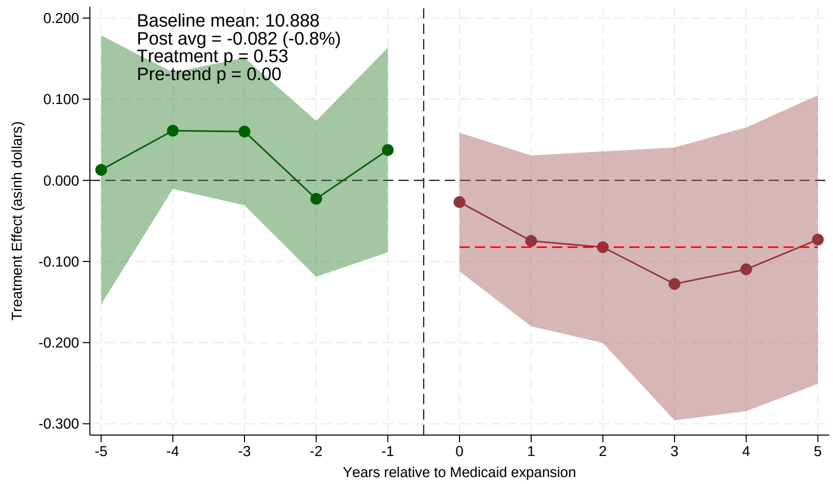

(a) Volume-Responsive GME States



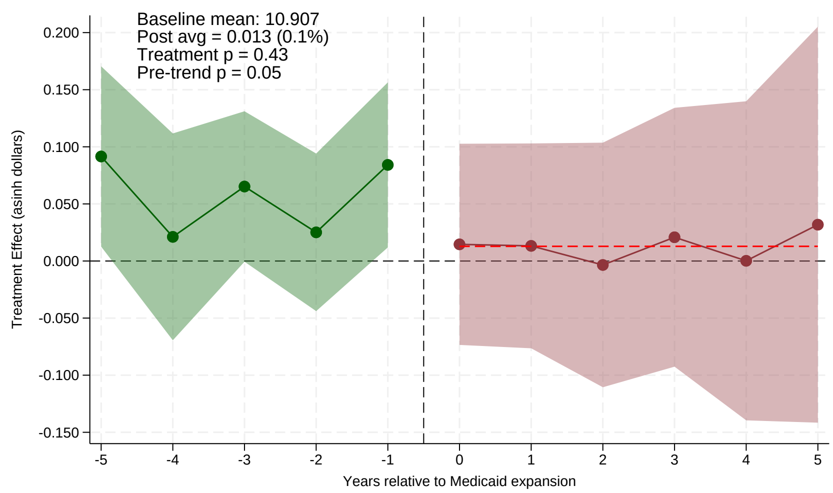

(b) Non-Responsive GME States (Fixed/None)

*Notes:* The figure decomposes the Direct GME payment response in Figure A.13: the outcome is the inverse hyperbolic sine of cost-report DGME resident FTEs per teaching hospital, so coefficients approximate proportional changes in the number of residents hospitals report for payment purposes. In volume-responsive states resident FTEs rise by roughly 19 percent (average post effect 0.186, s.e. 0.065; joint $p < 0.001$) after flat pre-trends ($p = 0.75$); in non-responsive states FTEs do not move (+3 percent, s.e. 0.055; joint $p = 0.95$; pre-trend $p = 0.81$). The cross-group difference is 0.14 (s.e. 0.05, $p = 0.005$ from a pooled interacted model). These estimates are on the full unweighted cost-report sample; on the linked NRMP–cost-report sample with the headline's weights and year convention the FTE response is negative in both arms (see Section 7.3 and the linked-sample reconciliation exhibits), so the unweighted rise here should not be read as evidence that training expanded. The DGME payment growth in volume-responsive states is therefore a quantity response, while the smaller payment rise in non-responsive states occurs without any growth in residents. Green circles (years $-5$ to $-1$) show pre-expansion coefficients; dark red circles (years 0 to +5) show post-expansion effects. Shaded bands are 95 percent confidence intervals. Estimates come from CMS Medicare hospital cost reports at the hospital–fiscal-year level with never-expansion states as the comparison group. Regressions include hospital and year fixed effects and are unweighted. Standard errors are clustered at the state level.

Figure A.16: Decomposition of the DGME Payment Response: Payment per Resident FTE by State Medicaid GME Formula



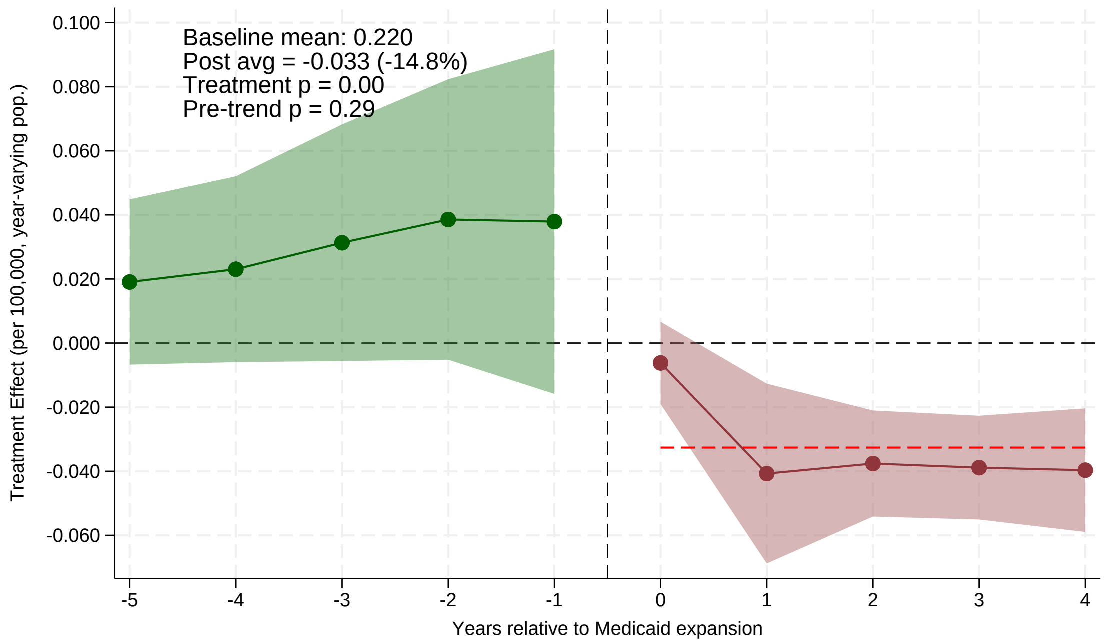

(a) Volume-Responsive GME States




(b) Non-Responsive GME States (Fixed/None)

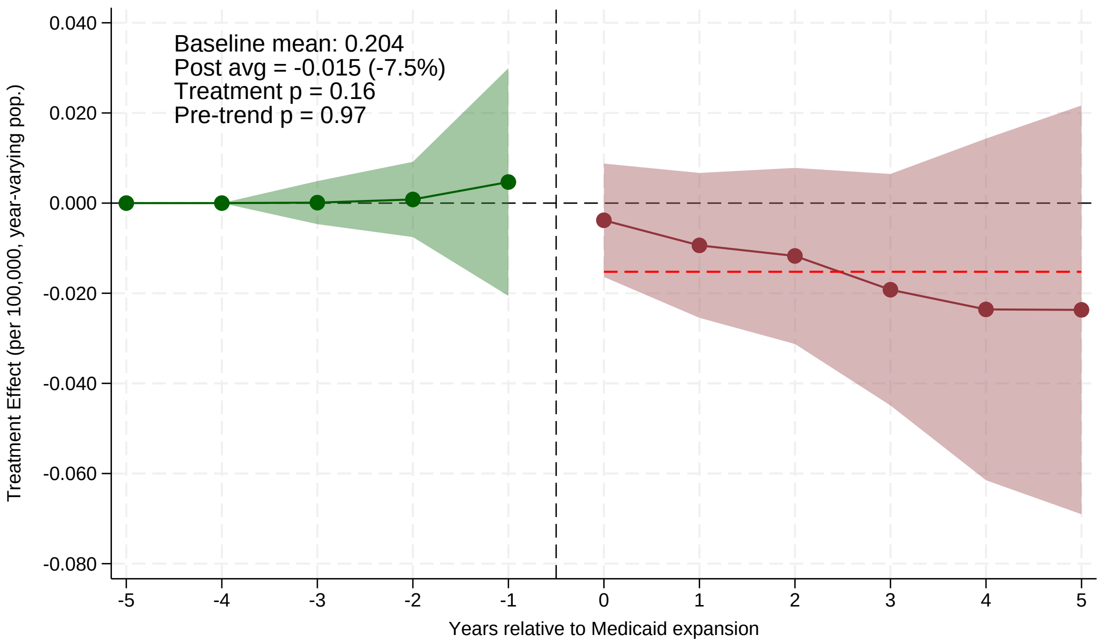
*Notes:* The figure completes the decomposition in Figure A.15: the outcome is the inverse hyperbolic sine of DGME payment per resident FTE (hospitals with positive FTEs only). Per-FTE payments do not rise in either group: the average post effect is $-0.08$ in volume-responsive states (s.e. 0.06; joint $p = 0.53$) and $+0.01$ in non-responsive states (s.e. 0.05; joint $p = 0.43$; pre-trend $p = 0.053$); the cross-group difference is $-0.09$ (s.e. 0.04, $p = 0.03$). The volume-responsive panel exhibits a differential pre-trend (pre-trend $p < 0.001$), so its level should be read with caution; the substantive conclusion—that the payment response is a quantity response, not a rate response—rests on the FTE panel. Green circles (years $-5$ to $-1$) show pre-expansion coefficients; dark red circles (years 0 to $+5$) show post-expansion effects. Shaded bands are 95 percent confidence intervals. Estimates come from CMS Medicare hospital cost reports at the hospital–fiscal-year level with never-expansion states as the comparison group. Regressions include hospital and year fixed effects and are unweighted. Standard errors are clustered at the state level.

Figure A.17: Robustness to Not-Yet-Treated Controls: Effect of Medicaid Expansion on Matched Residency Positions per 100,000 Population



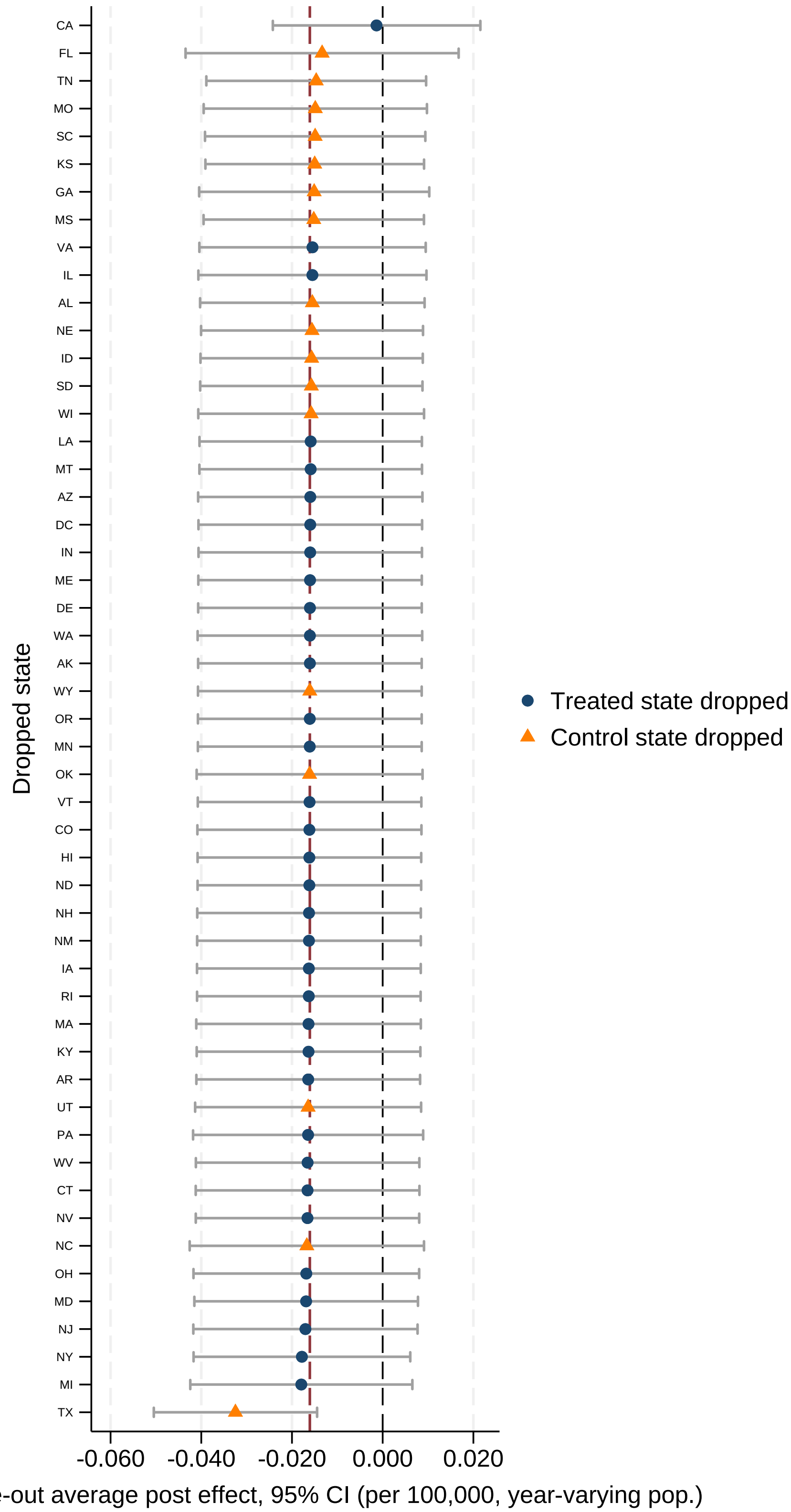

*Notes:* The figure re-estimates the main event study after dropping never-expansion states, so identification rests solely on variation in the timing of expansion among eventually-treated states (the not-yet-treated design). The outcome is matched positions per 100,000 contemporary (year-varying) population. Because the last-treated cohort has no valid comparison at horizon +5 under the not-yet-treated design, the event study is estimated over horizons 0 to +4. The average post-expansion effect is −0.033 (a 14.8 percent decline relative to the pre-expansion mean of 0.22; joint clustered $p < 0.01$; randomization-inference $p = 0.38$, see Section 7.3)—larger and, under clustered inference, more precisely estimated than the full-sample effect—with flat pre-expansion coefficients (pre-trend $p = 0.29$). The contraction is therefore not an artifact of the never-expansion comparison group; if anything it is sharper when identification rests solely on the timing of expansion among adopters. Green circles (years −5 to −1) show pre-expansion coefficients; dark red circles (years 0 to +4) show post-expansion effects; the dashed red line marks the average post-expansion effect. Shaded bands are 95 percent confidence intervals. Regressions include hospital and year fixed effects and weight by 2010 state population. Standard errors are clustered at the state level.

Figure A.18: Balanced-Cohort Event Study: 2014 Adopters Only



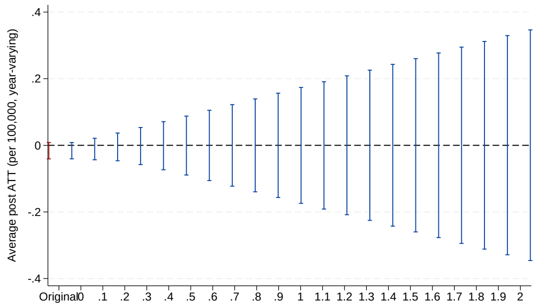

*Notes:* The figure re-estimates the main event study restricting treatment to the 2014 expansion cohort (the largest wave), with never-expansion states as the comparison group, so that every event-time coefficient is identified from the same fixed set of treated states. The outcome is matched positions per 100,000 contemporary (year-varying) population. The average post-expansion effect is $-0.015$ (a 7.5 percent decline relative to the pre-expansion mean of 0.20; joint $p = 0.16$ in this smaller single-cohort sample), and the event-time coefficients widen monotonically from $-0.004$ at horizon 0 to $-0.024$ at horizon $+5$, following flat pre-expansion coefficients (pre-trend $p = 0.97$). Because cohort composition is held fixed, the widening path here reflects genuine dynamics rather than the changing composition of cohorts across horizons. Green circles (years $-5$ to $-1$) show pre-expansion coefficients; dark red circles (years 0 to $+5$) show post-expansion effects; the dashed red line marks the average post-expansion effect. Shaded bands are 95 percent confidence intervals. Regressions include hospital and year fixed effects and weight by 2010 state population. Standard errors are clustered at the state level.

Figure A.19: Leave-One-State-Out Estimates of the Headline Effect

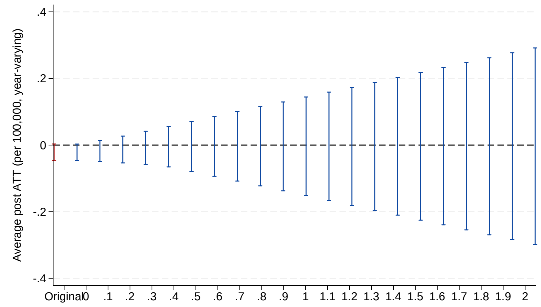



*Notes:* The figure plots the average post-expansion effect on matched positions per 100,000 contemporary (year-varying) population, re-estimated after dropping each treated state in turn (34 estimates, sorted). The dashed maroon line marks the full-sample estimate (−0.016); the thin black line marks zero. Every leave-one-out estimate remains negative (range −0.018 to −0.001), so the sign of the effect does not depend on any single state; dropping California alone, however, shrinks

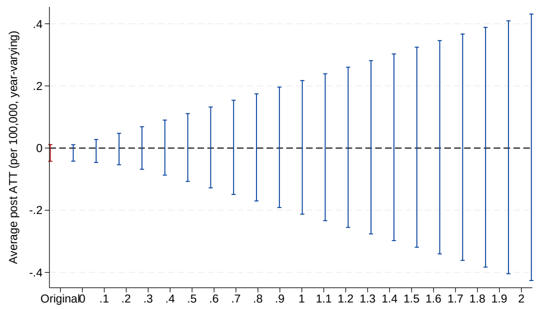
Figure A.20: HonestDiD Sensitivity to Violations of Parallel Trends




(a) Headline (per 100,000, year-varying)




(b) Non-Responsive States




(c) Urban Hospitals

*Notes:* The figure reports Rambachan and Roth (2023) sensitivity analysis for the average post-expansion effect on matched positions per 100,000 contemporary population, under the relative-magnitudes restriction $\Delta^{RM}(\bar{M})$, which bounds the post-period differential trend by $\bar{M}$ times the largest pre-period differential trend. Each panel plots the robust 95 percent confidence interval for the average post ATT as $\bar{M}$ increases. For the headline effect, the non-responsive mechanism panel, and the urban subsample, the robust confidence interval includes zero even at $\bar{M} = 0$ (exact parallel trends), confirming that the average magnitude is not distinguishable from zero at conventional levels; what the design supports is the sign and the growing dynamic path of the effect, together with the clustered joint tests and the not-yet-treated estimate. The leftmost interval in each panel is the original CI. Estimates use the Borusyak, Jaravel, and Spiess (2024) event-study coefficients with state-clustered standard errors.